# Notes on *Modern Quantum Mechanics* by Sakurai & Napolitano

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## 3.1 Rotation and Angular Momentum Commutation Relations

### 3.1.1 Finite Versus Infinitesimal Rotations

It is trivial that a 30-degree rotation about an axis followed by a 60-degree rotation about the same axis is equivalent to starting with 60- and ending with 30-degrees of rotation, thus rotations about the same axis commute. The same is not true for different axes of rotation, i.e. a 90° rotation about the $x$-axis followed by a 90° rotation about the $y$-axis is not equivalent to the reverse. The rotation of a vector $\mathbf{V}$, can be described by the orthogonal matrix $R_n(\varphi)$ such that a $\varphi$ rotation about the axis $n$ is given by

$$\mathbf{V}' = R_n(\varphi)\mathbf{V}. \tag{3.1}$$

The orthogonality of $R_n(\varphi)$ implies that $R_n^{-1} = R_n^T$ and

$$\sqrt{V_x^2 + V_y^2 + V_z^2} = \sqrt{V_x'^2 + V_y'^2 + V_z'^2}. \tag{3.2}$$

The convention used in these notes as well as the literature is that the coordinate system is static while the physical system rotates (active rotation), and positive rotation is defined as counterclockwise, i.e. the right-hand rule applies. For a rotation about the $z$-axis, it can be shown that

$$R_z(\varphi) = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.3}$$

If we want tot look at an infinitesimal rotation $\varphi = \varepsilon$, where $\varepsilon^3 = 0$, we can perform a Taylor expansion of the rotation matrix yielding

$$R_z(\varepsilon) = \begin{pmatrix} 1-\dfrac{\varepsilon^2}{2} & -\varepsilon & 0 \\ \varepsilon & 1-\dfrac{\varepsilon^2}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.4}$$

$$R_x(\varepsilon) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\dfrac{\varepsilon^2}{2} & -\varepsilon \\ 0 & \varepsilon & 1-\dfrac{\varepsilon^2}{2} \end{pmatrix} \tag{3.5}$$

$$R_y(\varepsilon) = \begin{pmatrix} 1-\dfrac{\varepsilon^2}{2} & 0 & \varepsilon \\ 0 & 1 & 0 \\ -\varepsilon & 0 & 1-\dfrac{\varepsilon^2}{2} \end{pmatrix}. \tag{3.6}$$

By performing two successive infinitesimal rotation about the $x$- and $y$-axis, we find that

$$R_x(\varepsilon)R_y(\varepsilon) = \begin{pmatrix} 1-\dfrac{\varepsilon^2}{2} & 0 & \varepsilon \\ \varepsilon^2 & 1-\dfrac{\varepsilon^2}{2} & -\varepsilon \\ -\varepsilon & \varepsilon & 1-\dfrac{\varepsilon^2}{2} \end{pmatrix} \tag{3.7}$$

$$R_y(\varepsilon)R_x(\varepsilon) = \begin{pmatrix} 1-\dfrac{\varepsilon^2}{2} & 0 & \varepsilon \\ -\varepsilon^2 & 1-\dfrac{\varepsilon^2}{2} & -\varepsilon \\ -\varepsilon & \varepsilon & 1-\dfrac{\varepsilon^2}{2} \end{pmatrix}, \tag{3.8}$$

which shows that the commutation relation

$$[R_x(\varepsilon), R_y(\varepsilon)] = \begin{pmatrix} 0 & -\varepsilon^2 & 0 \\ \varepsilon^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\varepsilon^2) - 1 = R_z(\varepsilon^2) - R_n(0), \tag{3.9}$$

notably we find that the rotation commute if we ignore terms of order $\varepsilon^2$ and higher.

### 3.1.2 Infinitesimal Rotations in Quantum Mechanics

The previous subsection did not involve quantum mechanical concepts, which we will introduce now. Since a rotation is active, we change the system, and thus the state ket after a rotation should be different from the original one. For a rotation characterized by the matrix $R$, we introduce the operator $\mathscr{D}(R)$ such that

$$\alpha_R = \mathscr{D}(R)\alpha. \tag{3.10}$$

Note that the matrix representation of $\mathscr{D}(R)$ is not the same as $R$, but is instead dependent on the dimensionality of the ket-space of the system, i.e. for a spin $\frac{1}{2}$ system we have a $2 \times 2$ matrix while for a spin 1 system we have a $3 \times 3$ matrix representation of $\mathscr{D}(R)$.

To construct this operator we start in the same way as for the construction of a translation and time evolution operator by looking at an infinitesimal rotation

$$U_\varepsilon = 1 - iG\varepsilon, \tag{3.11}$$

where $G$ is a generator of the rotation. From classical mechanics we know that the generator of rotation is angular momentum, and we thus have

$$G \to \frac{\mathbf{J}\cdot\hat{\mathbf{n}}}{\hbar}, \quad \varepsilon \to d\varphi, \tag{3.12}$$

where $\hat{\mathbf{n}}$ is the axis of rotation, and we rotate around this axis by an angle $d\varphi$, and we note that the angular momentum operator $\mathbf{J}$ is defined by this expression. Taking $\mathbf{J}$ to be Hermitian we guarantee $\mathscr{D}(R)$ to be unitary. We can thus write the operator for an infinitesimal rotation as

$$\mathscr{D}(\hat{\mathbf{n}}, d\varphi) = 1 - \frac{i}{\hbar}\mathbf{J}\cdot\hat{\mathbf{n}}d\varphi. \tag{3.13}$$

Note that this formalism is applicable for any type of angular momentum, and we therefore do not take $\mathbf{J}$ to be defined by $\mathbf{x}\times\mathbf{p}$ as in classical mechanics, but instead let $J$ be defined such that the above expression is true for an infinitesimal rotation.

### 3.1.3 Finite Rotations in Quantum Mechanics

If we want to consider a finite rotation we perform $N$ successive infinitesimal rotations of the form in Eq. (3.13). For a rotation about the $z$-axis we have

$$\begin{aligned}\mathscr{D}(\hat{\mathbf{z}},\varphi) &= \lim_{N\to\infty}\left[1-\frac{i}{\hbar}J_z\frac{\varphi}{N}\right]^N && (3.14)\\ &= 1-\frac{iJ_z\varphi}{\hbar}+\frac{J_z^2\varphi^2}{2\hbar^2}+\cdots && (3.15)\\ &= \exp\left(-\frac{iJ_z\varphi}{\hbar}\right). && (3.16)\end{aligned}$$

We postulate that $\mathscr{D}$ has the same group properties as $R$, which are

$$\begin{array}{lrcll} \text{Identity:} & R\cdot 1 = R & \Longrightarrow & \mathscr{D}(R)\cdot 1 = \mathscr{D}(R) & (3.17a)\\ \text{Closure:} & R_1R_2 = R_3 & \Longrightarrow & \mathscr{D}(R_1)\mathscr{D}(R_2) = \mathscr{D}(R_3) & (3.17b)\\ \text{Inverses:} & RR^{-1} = 1 & \Longrightarrow & \mathscr{D}(R)\mathscr{D}^{-1}(R) = 1 & (3.17c)\\ & R^{-1}R = 1 & \Longrightarrow & \mathscr{D}^{-1}(R)\mathscr{D}(R) = 1 & (3.17d)\\ \text{Associativity:} & R_1(R_2R_3) = (R_1R_2)R_3 & \Longrightarrow & \mathscr{D}(R_1)[\mathscr{D}(R_2)\mathscr{D}(R_3)] = [\mathscr{D}_1\mathscr{D}_2]\mathscr{D}_3. & (3.17e)\end{array}$$

### 3.1.4 Commutation Relations for Angular Momentum

The fundamental commutation relation for angular momentum is

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k, \tag{3.18}$$

where $\varepsilon_{ijk}$ is the Levi-Civita symbol. When generators of infinitesimal transformations (like rotation) do not commute we call the corresponding group non-Abelian. Rotation is thus non-Abelian while translation is Abelian since $[p_i, p_j] = 0$. It is important to emphasize that to obtain the commutation relations for angular momentum we use that $J_k$ is the rotation about the $k$-axis, and that rotation about different axes do not commute. Simply put, The commutation relation Eq. (3.18) summarizes all basic properties of rotations in three dimensions.

To see how the system rotates we can look at the expectation value of an operator in a state $\alpha$ and its rotated version $\alpha_R = \mathscr{D}_z(\varphi)\alpha$, which gives

$$\langle J_x\rangle = {}_R\alpha J_x\alpha_R = \alpha\mathscr{D}_z^\dagger(\varphi)J_x\mathscr{D}_z(\varphi)\alpha, \tag{3.19}$$

we thus need to compute

$$\mathscr{D}_z^\dagger(\varphi)J_x\mathscr{D}_z(\varphi) = \exp\left(\frac{iJ_z\varphi}{\hbar}\right)J_x\exp\left(-\frac{iJ_z\varphi}{\hbar}\right) = J_x\cos\varphi - J_y\sin\varphi, \tag{3.20}$$

and we thus see that we change the expectation value of $J_x$ by rotating the system.

## 3.2 Spin $\frac{1}{2}$ Systems and Finite Rotations

Experimental evidence suggests that nature realizes the lowest possible dimensional realization of angular momentum, which for a spin $\frac{1}{2}$ system is $N = 2$. We can write the spin angular momentum operators as

$$S_x = \frac{\hbar}{2}(|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|) = \frac{\hbar}{2}\sigma_x \tag{3.21a}$$
$$S_y = \frac{i\hbar}{2}(-|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|) = \frac{i\hbar}{2}\sigma_y \tag{3.21b}$$
$$S_z = \frac{\hbar}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|) = \frac{\hbar}{2}\sigma_z. \tag{3.21c}$$

As shown in the previous subsection the behaviour of a rotation on the expectation value is given by

$$\langle S_x\rangle_R = \langle S_x\rangle\cos\varphi - \langle S_y\rangle\sin\varphi \tag{3.22a}$$
$$\langle S_y\rangle_R = \langle S_y\rangle\sin\varphi + \langle S_x\rangle\cos\varphi \tag{3.22b}$$
$$\langle S_z\rangle_R = \langle S_z\rangle, \tag{3.22c}$$

where we do not see a change in $\langle S_z\rangle$ since $S_z$ commutes with the rotation operator $\mathscr{D}_z(\varphi)$. We see that the behaviour is that of a classical vector rotating in the $xy$-plane, and is general of any $J_k$ and not just for spin.

Examining the rotation operator acting on a general ket

$$\alpha = \langle\uparrow|\alpha\rangle\uparrow + \langle\downarrow|\alpha\rangle\downarrow, \tag{3.23}$$

we find

$$\mathscr{D}_z(\varphi)\alpha = e^{-i\varphi/2}\langle\uparrow|\alpha\rangle\uparrow + e^{i\varphi/2}\langle\downarrow|\alpha\rangle\downarrow, \tag{3.24}$$

which has the interesting consequence (due to the half-angle) that a full $2\pi$-rotation does not return the system to its original state,

$$\mathscr{D}_z(2\pi)\alpha = -\alpha, \tag{3.25}$$

such that we need a $4\pi$-rotation. For the calculation of an expectation value, both the bra and ket are rotated the same amount and thus the mins signs cancels. The question is therefore if this is possible to observe experimentally?

### 3.2.1 Spin Precession Revisited

For a magnetic field $\mathbf{B} = (0, 0, B_z)^T$ and the spin interaction Hamiltonian

$$H = -\frac{e}{m_e c}\mathbf{S}\cdot\mathbf{B} = \omega S_z, \quad \omega = \frac{|e|B_z}{m_e c}, \tag{3.26}$$

where we recall that $e < 0$ for the electron. The time evolution operator for this Hamiltonian is given by

$$\mathscr{U}(t, 0) = \exp\left(-\frac{iHt}{\hbar}\right) = \exp\left(-\frac{iS_z\omega t}{\hbar}\right), \tag{3.27}$$

which is the rotation operator $\mathscr{D}_z(\omega t)$, and it is clear why this Hamiltonian causes spin precession. By using the Eqs. (3.22) and (3.24) we find that the period for the precession is $\tau_{\text{precession}} = 2\pi/\omega$ and for the state ket $\tau_{\text{state ket}} = 4\pi/\omega$.

## 3.3 SO(3), SU(2), and Euler Rotations

### 3.3.1 Orthogonal Group

The simplest way to characterize a rotation is by defined the axis of rotation ($\hat{\mathbf{n}}$) and the angle rotated ($\varphi$). This requires three numbers, the azimuthal and polar angle of $\hat{\mathbf{n}}$ as well as the angle rotated $\varphi$. This is however not a good way to characterize the rotations when we want to study the group properties. Instead, we characterize the rotation with the rotation matrix $R$. Using the orthogonality condition $RR^T = R^T R = 1$, we find 6 independent equations which reduce the numbers needed to describe the matrix to $9 - 6 = 3$, which is the same number as the simple way.

We say that the set of all multiplication operations with orthogonal matrices form a group, which mean that the following conditions are satisfied

1. The product of any two orthogonal matrices is another orthogonal matrix, which is satisfied because
$$(R_1R_2)(R_1R_2)^T = R_1R_2R_2^T R_1^T = 1. \tag{3.28}$$

2. The associative law
$$R_1(R_2R_3) = (R_1R_2)R_3 \tag{3.29}$$
holds.

3. The identity matrix, physically corresponding to no rotation, is a member of the class of all orthogonal matrices.

4. The inverse matrix, physically corresponding to a rotation in the reverse direction, is also a member.

This group is called Special Orthogonal group of dimension 3, or SO(3). The name special is due to only including rotation operations, and not for example the inverse operation.

### 3.3.2 Unitary Unimodular Group

We can characterize a rotation by a $2 \times 2$ matrix acting on a spinor

$$\chi = \begin{pmatrix} \langle\uparrow|\alpha\rangle \\ \langle\downarrow|\alpha\rangle \end{pmatrix} = \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = c_+\chi_+ + c_-\chi_- \tag{3.30}$$

with $|c_+|^2 + |c_-|^2 = 1$. The most general unitary unimodular matrix can be written as

$$U(a,b) = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \tag{3.31}$$

where $a, b \in \mathbb{C}$ and satisfy the unimodular condition

$$|a|^2 + |b|^2 = 1, \tag{3.32}$$

and the matrix fulfils the unitarity condition

$$U^\dagger(a,b)U(a,b) = 1. \tag{3.33}$$

This matrix can be used to describe a rotation of a spin $\frac{1}{2}$ system.

The complex parameters are called the Cayley-Klein parameters, and is not quantum mechanical but was used in classical mechanics of rotation before the birth of quantum mechanics.

Other properties of unitary unimodular matrices are

$$U(a_1, b_1)U(a_2, b_2) = U(a_1a_2 - b_1b_2^*, a_1b_2 + a_2^*b_1) \tag{3.34}$$

fulfilling the unimodular condition, and the inverse

$$U^-1(a, b) = U(a^*, -b). \tag{3.35}$$

This group is called Special Unitary group of dimensionality 2, or SU(2). A general Unitary matrix can be written as

$$U = e^{i\gamma}\begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1, \quad \gamma^* = \gamma. \tag{3.36}$$

Even though both SO(3) and SU(2) can be used to characterize a rotation, they are not isomorphic, since SO(3) have a period of $2\pi$ while SU(2) have a period of $4\pi$ with a $2\pi$ period being equal to minus the identity matrix. One can also say that both $U(a, b)$ and $U(-a, -b)$ map to the same rotation in SO(3).

### 3.3.3 Euler Rotations

The Euler rotations is a way to characterize rotation by three angles called Euler angles, which is commonly used in rigid-body classical mechanics. The idea of Euler rotations is to rotate about the $z$-axis, then the $y$-axis, and once again the $z$-axis. Conventionally, in classical mechanics you change the $y$-axis for the $x$-axis, as well as rotate about the body axes instead of the fixed axes. This total rotation may be described as

$$R(\alpha, \beta, \gamma) = R_z(\alpha)R_y(\beta)R_z(\gamma). \tag{3.37}$$

There is a correspondence to the above relation but for rotation operators, which is

$$\mathscr{D}(\alpha, \beta, \gamma) = \mathscr{D}_z(\alpha)\mathscr{D}_y(\beta)\mathscr{D}_z(\gamma) \tag{3.38}$$

acting on a ket-space. For a spin $\frac{1}{2}$ system we get

$$\begin{align} \mathscr{D}(\alpha, \beta, \gamma) &= \exp\left(-\frac{i\sigma_3\alpha}{2}\right)\exp\left(-\frac{i\sigma_2\beta}{2}\right)\exp\left(-\frac{i\sigma_3\gamma}{2}\right) \tag{3.39} \\ &= \begin{pmatrix} e^{-i(\alpha+\gamma)/2}\cos(\beta/2) & -e^{-i(\alpha-\gamma)/2}\sin(\beta/2) \\ e^{i(\alpha-\gamma)/2}\sin(\beta/2) & e^{i(\alpha+\gamma)/2}\cos(\beta/2) \end{pmatrix}, \tag{3.40} \end{align}$$

which is a SU(2) matrix. This matrix is called the $J = 1/2$ irreducible representation of rotation operator $\mathscr{D}(\alpha, \beta, \gamma)$, and we have the matrix elements

$$\mathscr{D}^{1/2}_{m'm}(\alpha, \beta, \gamma) = \left\langle j = 1/2, m' \middle| \exp\left(-\frac{iJ_z\alpha}{\hbar}\right)\exp\left(-\frac{iJ_y\beta}{\hbar}\right)\exp\left(-\frac{iJ_z\gamma}{\hbar}\right) \middle| j = 1/2, m \right\rangle \tag{3.41}$$

## 3.4 Density Operators and Pure Versus Mixed Ensembles

### 3.4.1 Polarized Versus Unpolarized Beams

Up to this point the formalism deals with an ensemble of identically prepared kets $|\alpha\rangle$. We have however not dealt with situations where, say 60 % of the ensemble is prepared in $|\alpha\rangle$ and 40 % in $|\beta\rangle$, with $\alpha \neq \beta$. To solve this we introduce the concept of fractional population with probability weights, where we have some number $w_\uparrow$ being the probability of the system being prepared in the state $|\uparrow\rangle$ as well as $w_\downarrow = 1 - w_\uparrow$. Note that these are real numbers that lack phase information and are thus inherently different from a quantum mechanical superposition and instead is related to classical probability theory.

If an ensemble is completely randomized, it is called an unpolarized beam, which in the spin $\frac{1}{2}$ example would be $w_\uparrow = w_\downarrow = 0.5$. If we have a pure ensemble it would mean that we only have one spin orientation and thus $w_\uparrow = 1$ or $w_\downarrow = 1$, and we call this for a polarized beam. We could also imagine having something in between these two extremes, which is what we call a mixed ensemble or a partially polarized beam. Note that the different kets need not be orthogonal.

### 3.4.2 Ensemble Averages and Density Operator

To solve the problem of not just dealing with pure ensembles we introduce the density matrix formalism, which is a general formalism valid for any type of quantum mechanical system, but we will keep the examples in the spin $\frac{1}{2}$ system. We will consider an ensemble where a $w_1$ fraction of the system is in state $\left|\alpha^{(1)}\right\rangle$, a $w_1$ fraction is in state $\left|\alpha^{(2)}\right\rangle$, and so on, with the normalization condition

$$\sum_i w_i = 1, \tag{3.42}$$

and there is no requirement on orthogonality for the kets. There is also no limitation of the number of terms $i$ regarding the dimensionality of the system. Given an operator $A$ with eigenkets $|a'\rangle$, the ensemble average is

$$[A] = \sum_i w_i \left\langle\alpha^{(i)}\middle|A\middle|\alpha^{(i)}\right\rangle = \sum_i \sum_{a'} w_i \left|\left\langle a'\middle|\alpha^{(i)}\right\rangle\right|^2 a'. \tag{3.43}$$

Note that there are two types of probability concepts here. First we have the quantum mechanical probability of the expectation value of $A$ with respect to $\left|\alpha^{(i)}\right\rangle$ and then a classical probability of the system being in that state. Taking this and expanding into a general basis $\{|b'\rangle\}$ we get

$$[A] = \sum_i w_i \sum_{b'b''} \left\langle\alpha^{(i)}\middle|b'\right\rangle \langle b'|A|b''\rangle \left\langle b''\middle|\alpha^{(i)}\right\rangle \tag{3.44}$$

$$= \sum_{b'b''} \left( \sum_i w_i \left\langle b''\middle|\alpha^{(i)}\right\rangle \left\langle\alpha^{(i)}\middle|b'\right\rangle \right) \langle b'|A|b''\rangle, \tag{3.45}$$

where the number of terms in the $b'$ and $b''$ sums are equal to the dimensionality of the ket space. We see that we have a prefactor in front of the factor dependent on $A$. Thus, we introduce the density operator

$$\rho = \sum_i w_i \left|\alpha^{(i)}\right\rangle \left\langle\alpha^{(i)}\right|. \tag{3.46}$$

We obtain the matrix elements of this using

$$\langle b''|\rho|b'\rangle = \sum_i w_i \langle b''|\alpha^{(i)}\rangle\langle\alpha^{(i)}|b'\rangle \tag{3.47}$$

with which we can return to the equation of the ensemble average and obtain

$$[A] = \sum_{b'b''}\langle b''|\rho|b'\rangle\langle b'|A|b''\rangle = \sum_{b''}\langle b''|\rho A|b''\rangle = \mathrm{Tr}\{\rho A\}. \tag{3.48}$$

Two important properties of the density matrix is that it is Hermitian and that

$$\mathrm{Tr}\{\rho\} = 1. \tag{3.49}$$

For a spin $\frac{1}{2}$ system we have dimensionality $N = 2$ and thus the density matrix is a $2 \times 2$ matrix. The imposed conditions mean that three independent real parameters are needed to characterize the density matrix. For this system the needed numbers are $[S_x]$, $[S_y]$, and $[S_z]$. Note that a mixed ensemble may be decomposed into pure ensembles in multiple ways, and there is no uniqueness. For a pure ensemble, meaning that there exist a $w_i = 1$ have the property

$$\rho^2 = \rho, \tag{3.50}$$

and

$$\mathrm{Tr}\{\rho^2\} = 1, \tag{3.51}$$

and the eigenvalues are either 0 or 1. For a mixed ensemble we always have

$$0 < \mathrm{Tr}\{\rho^2\} < 1. \tag{3.52}$$

### 3.4.3 Time Evolution of Ensembles

Let us now consider the time evolution of the density operator. Barring external interaction, we cannot change the relative weights $w_i$, but the kets $|\alpha^{(i)}\rangle$ will evolve in time according to the Schrödinger equation. We find from the Schrödinger equation that

$$i\hbar\partial_t\rho = -[\rho, H], \tag{3.53}$$

which looks like the Heisenberg equation of motion but with a sign flip. Important to note however, is that the Heisenberg equation of motion deals with the time evolution of operators, while the equation of motion for the density operator is derived from the Schrödinger picture and is therefore in the Schrödinger picture and not in the Heisenberg picture, even though it is an operator. We also note that this is a quantum mechanical analogue of Liouville's theorem in classical mechanics

$$\partial_t\rho_{\text{classical}} = -[\rho_{\text{classical}}, H]_{\text{classical}} \tag{3.54}$$

### 3.4.4 Continuum Generalizations

It is possible to generalize this formulation for ket spaces spanned by continuous variables. Consider the ket space spanned by the position eigenkets $|\mathbf{x}'\rangle$. We find that the ensemble average is given by

$$[A] = \int d^3x' \int d^3x'' \langle\mathbf{x}''|\rho|\mathbf{x}'\rangle\langle\mathbf{x}'|A|\mathbf{x}''\rangle, \tag{3.55}$$

and we find that the density matrix is described with

$$\langle \mathbf{x}''|\rho|\mathbf{x}'\rangle = \langle \mathbf{x}''|\left(\sum_i w_i \big|\alpha^{(i)}\big\rangle\big\langle\alpha^{(i)}\big|\right)|\mathbf{x}'\rangle = \sum_i w_i \psi_i(\mathbf{x}'')\psi_i^*(\mathbf{x}') \tag{3.56}$$

where the wave function appears. Note that the diagonal elements is the weighted sum of probability densities.

### 3.4.5 Quantum Statistical Mechanics

We will go through some properties of completely random and completely pure density matrices. Firstly, a completely random $N$-dimensional density matrix has the form

$$\rho = \frac{1}{N}\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}, \tag{3.57}$$

while a pure density matrix has the form

$$\rho = \begin{pmatrix} 0 & \dots & & & \dots & 0 \\ \vdots & \ddots & & & & \vdots \\ & & 0 & & & \\ & & & 1 & & \\ & & & & 0 & \\ \vdots & & & & & \ddots \\ 0 & \dots & & & \dots & 0 \end{pmatrix}. \tag{3.58}$$

These two matrices are very different, and we introduce the quantity

$$\sigma = -\mathrm{Tr}\{\rho \ln \rho\} \tag{3.59}$$

that characterizes this difference. Writing this quantity in the diagonal basis yields

$$\sigma = -\sum_k \rho_{kk}^{\text{diagonal}} \ln \rho_{kk}^{\text{diagonal}}, \tag{3.60}$$

and it follows from this that $\sigma$ must be positive semidefinite, since $0 < \rho_k k < 1$ for all $k$. For a completely random density matrix we find

$$\sigma = -\sum_{k=1}^{N} \frac{1}{N} \ln\left(\frac{1}{N}\right) = \ln N, \tag{3.61}$$

while for a pure ensemble we find

$$\sigma = 0. \tag{3.62}$$

We say that this is a measure of the disorder for a system, and that $\ln N$ is the maximum value this disorder can take for a density matrix under the normalization constraint. We find that $\sigma$ is related to the entropy per constituent member, $S$, by the relation

$$S = k_B \sigma, \tag{3.63}$$

where $k_b$ is the Boltzmann constant. Furthermore, we take this to be the definition of entropy in quantum statistical mechanics.

We make the assumption that nature wants to maximize this entropy, under the constraint that $[H]$ has a certain value. With thermal equilibrium we have

$$\partial_t \rho = 0, \tag{3.64}$$

which means that $\rho$ and $H$ commute, via the von Neumann equation of motion, and thus may both be diagonalized. That is, we can write $\rho$ in the basis of the energy eigenkets. Using calculus of variations we find that

$$\rho_{kk} = \frac{\exp(-\beta E_k)}{\sum_l^N \exp(-\beta E_l)}, \tag{3.65}$$

giving the fractional population of an energy eigenstate, where

$$\beta = \frac{1}{k_B T}. \tag{3.66}$$

This is known in statistical mechanics as the canonical ensemble. A completely random ensemble is the limit of $\beta \to 0$, or the high temperature limit. We also note that the denominator is the partition function, $Z$, from statistical mechanics. Thus, if we have $\rho$ in the energy eigenbasis we can write

$$\rho = \frac{e^{-\beta H}}{Z}. \tag{3.67}$$

This can then be used to calculate the ensemble average

$$[A] = \frac{\mathrm{Tr}\{e^{-\beta H} A\}}{Z} = \frac{\sum_k^N \langle A \rangle_k \exp(-\beta E_k)}{\sum_k^N \exp(-\beta E_k)}. \tag{3.68}$$

Calculating the internal energy per constituent gives

$$U = [H] = \frac{\sum_k^N E_k \exp(-\beta E_k)}{\sum_k^N \exp(-\beta E_k)} = -\partial_\beta \ln Z, \tag{3.69}$$

which is a well known formula in statistical mechanics. In the low temperature limit of $\beta \to \infty$ we obtain the pure ensemble, with only the ground state populated.

## 3.6 Orbital Angular Momentum

### 3.6.1 Orbital Angular Momentum as Rotation Generator

In the previous section we defined angular momentum as the generator of infinitesimal rotation. If the system lacks spin we can also define the angular momentum in terms of the orbital angular momentum

$$\mathbf{L} = \mathbf{x} \times \mathbf{p}, \tag{3.70}$$

which we will discuss in this section.

This definition of orbital angular momentum stasifies the known angular momentum commutation relations

$$[L_i, L_j] = i\varepsilon_{ijk}\hbar L_k, \tag{3.71}$$

which can be proven from the commutation relation between $\mathbf{x}$ and $\mathbf{p}$. We have $L_z = xp_y - yp_x$, and we study if this can be used to think about this as a generator of rotation by using the fact that momentum is the generator of translation. Thus,

$$\left[1 - i\frac{\delta\varphi}{\hbar}L_z\right]|\mathbf{x}'\rangle = \left[1 - i\frac{p_y}{\hbar}\delta\varphi x' + i\frac{p_x}{\hbar}\delta\varphi y'\right]|\mathbf{x}'\rangle \tag{3.72}$$

$$= |x' - y'\delta\varphi, y' + x'\delta\varphi, z'\rangle, \tag{3.73}$$

which is an infinitesimal rotation around the $z$-axis. If the momentum operator is a generator of translation, we have shown that the orbital angular momentum operator is a generator of rotation. Switching to spherical coordinates, and suppose we have a wave function $\langle\mathbf{x}|\alpha\rangle$ for an arbitrary system ket $|\alpha\rangle$ without spin, we can find the rotated wave function

$$\langle\mathbf{x}'|\left[1 - i\frac{\delta\varphi}{\hbar}L_z\right]|\alpha\rangle = \langle r, \vartheta, \varphi - \delta\varphi|\alpha\rangle \tag{3.74}$$

$$= \langle\mathbf{x}'|\alpha\rangle - \delta\varphi\partial_\varphi\langle\mathbf{x}'|\alpha\rangle, \tag{3.75}$$

from which we can identify

$$\langle\mathbf{x}'|L_z|\alpha\rangle = -i\hbar\partial_\varphi\langle\mathbf{x}'|\alpha\rangle. \tag{3.76}$$

Considering rotation about the $x$- and $y$-axes we find

$$\langle\mathbf{x}'|L_x|\alpha\rangle = -i\hbar(-\sin\varphi\partial_\vartheta - \cot\vartheta\cos\varphi\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle \tag{3.77}$$

$$\langle\mathbf{x}'|L_y|\alpha\rangle = -i\hbar(\cos\varphi\partial\vartheta - \cot\vartheta\sin\varphi\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle. \tag{3.78}$$

We can define the ladder operators $L_\pm$ using these, and we find

$$\langle\mathbf{x}'|L_\pm|\alpha\rangle = -i\hbar e^{\pm i\varphi}(\pm i\partial_\vartheta - \cot\vartheta\partial_\varphi)\langle\mathbf{x}'|\alpha\rangle. \tag{3.79}$$

Using

$$\mathbf{L}^2 = L_z^2 + \frac{1}{2}(L_+L_- + L_-L_+), \tag{3.80}$$

we also find

$$\left\langle\mathbf{x}'\middle|\mathbf{L}^2\middle|\alpha\right\rangle = -\hbar^2\left[\frac{1}{\sin^2\vartheta}\partial_\varphi^2 + \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta)\right]\langle\mathbf{x}'|\alpha\rangle, \tag{3.81}$$

which, apart from the factor $1/r^2$, is the angular part of the Laplacian $\nabla^2 = \Delta$ in spherical coordinates. It is possible to show this from the kinetic energy operator as well. We start by writing the operator identity

$$\mathbf{L}^2 = \mathbf{x}^2\mathbf{p}^2 - (\mathbf{x}\cdot\mathbf{p})^2 + i\hbar\mathbf{x}\cdot\mathbf{p}, \tag{3.82}$$

and we also have

$$\langle\mathbf{x}'|\mathbf{x}\cdot\mathbf{p}|\alpha\rangle = -i\hbar r\partial_r\langle\mathbf{x}'|\alpha\rangle \tag{3.83}$$

$$\left\langle\mathbf{x}'\middle|(\mathbf{x}\cdot\mathbf{p})^2\middle|\alpha\right\rangle = -\hbar^2(r^2\partial_r^2\langle\mathbf{x}'|\alpha\rangle + r\partial_r\langle\mathbf{x}'|\alpha\rangle). \tag{3.84}$$

Using these we find

$$\left\langle\mathbf{x}'\middle|\mathbf{L}^2\middle|\alpha\right\rangle = r^2\left\langle\mathbf{x}'\middle|\mathbf{p}^2\middle|\alpha\right\rangle + \hbar(r^2\partial_r^2\langle\mathbf{x}'|\alpha\rangle + 2r\partial_r\langle\mathbf{x}'|\alpha\rangle). \tag{3.85}$$

Rewriting such that we get the kinetic energy operator $\mathbf{p}^2/2m$, we get

$$\frac{1}{2m}\big\langle\mathbf{x}'\big|\mathbf{p}^2\big|\alpha\big\rangle = -\frac{\hbar^2}{2m}\left(\partial_r^2\langle\mathbf{x}'|\alpha\rangle + \frac{2}{r}\partial_r\langle\mathbf{x}'|\alpha\rangle - \frac{1}{\hbar^2 r^2}\big\langle\mathbf{x}'\big|\mathbf{L}^2\big|\alpha\big\rangle\right), \tag{3.86}$$

and since we have

$$\frac{1}{2m}\big\langle\mathbf{x}'\big|\mathbf{p}^2\big|\alpha\big\rangle = -\frac{\hbar^2}{2m}\nabla'^2\langle\mathbf{x}'|\alpha\rangle, \tag{3.87}$$

we can identify $\big\langle\mathbf{x}'\big|\mathbf{L}^2\big|\alpha\big\rangle$ as the angular part of the Laplacian.

### 3.6.2 Spherical Harmonics

We consider a spinless particle in a spherical symmetrical potential. The wave function is separable in spherical coordinates, and we can write the energy eigenfunctions as

$$\langle\mathbf{x}'|n, l, m\rangle = R_{nl}(r)Y_l^m(\vartheta, \varphi), \tag{3.88}$$

which is a direct consequence of the rotational invariance of the system. For a spherical symmetric system $H$ commutes with $L_z$ and $\mathbf{L}^2$ and the energy eigenkets are thus eigenkets for $L_z$ and $\mathbf{L}^2$ also. Since the angular dependence is the same for all systems with spherical symmetry we can isolate this using the direction eigenket $|\hat{\mathbf{n}}\rangle$, and write

$$\langle\hat{\mathbf{n}}|l, m\rangle = Y_l^m(\vartheta, \varphi) = Y_l^m(\hat{\mathbf{n}}). \tag{3.89}$$

We can think of $Y_l^m(\vartheta, \varphi)$ as the amplitude of the state characterized by $l, m$ found in the direction of $\hat{\mathbf{n}}$ specified by the angles $\vartheta, \varphi$. From eigenvalue equations such as

$$L_z|l, m\rangle = m\hbar|l, m\rangle \tag{3.90}$$

we can infer the dependence on $Y_l^m(\vartheta, \varphi)$. Multiplying by $\langle\hat{\mathbf{n}}|$ we find

$$\langle\hat{\mathbf{n}}|L_z|l, m\rangle = -i\hbar\partial_\varphi\langle\hat{\mathbf{n}}|l, m\rangle = m\hbar\langle\hat{\mathbf{n}}|l, m\rangle, \tag{3.91}$$

which is

$$-i\hbar\partial\varphi Y_l^m(\vartheta, \varphi) = m\hbar Y_l^m(\vartheta, \varphi), \tag{3.92}$$

and thus the dependence on $\varphi$ of $Y_l^m(\vartheta, \varphi)$ must be $e^{im\varphi}$. Similarly, using the eigenvalue equation

$$\mathbf{L}^2|l, m\rangle = l(l+1)\hbar^2|l, m\rangle \tag{3.93}$$

we have

$$\big\langle\hat{\mathbf{n}}\big|\mathbf{L}^2\big|l, m\big\rangle = l(l+1)\hbar^2\langle\hat{\mathbf{n}}|l, m\rangle = -\hbar^2\left[\frac{1}{\sin^2\vartheta}\partial_\varphi^2 + \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta)\right]\langle\hat{\mathbf{n}}|l, m\rangle, \tag{3.94}$$

which is equivalent to

$$\left[\frac{1}{\sin\vartheta}\partial_\vartheta(\sin_\vartheta\partial\vartheta) + \frac{1}{\sin^2\vartheta}\partial_\varphi^2 + l(l+1)\right]Y_l^m(\vartheta, \varphi) = 0. \tag{3.95}$$

The orthogonality relation

$$\langle l', m'|l, m\rangle = \delta_{ll'}\delta_{mm'} \tag{3.96}$$

is equivalent to

$$\int_0^{2\pi} d\varphi \int_{-1}^{1} d\cos\vartheta Y_{l'}^{m'*}(\vartheta,\varphi) Y_l^m(\vartheta,\varphi) = \delta_{ll'}\delta_{mm'}. \tag{3.97}$$

Consider the case $m = l$. We have the equation

$$L_+|l,l\rangle = 0, \tag{3.98}$$

which leads to

$$-i\hbar e^{i\varphi}[i\partial_\varphi - \cot\vartheta\partial_\varphi]\langle\hat{\mathbf{n}}|l,l\rangle = 0, \tag{3.99}$$

and this partial differential equation is satisfied by

$$\langle\hat{\mathbf{n}}|l,l\rangle = Y_l^l(\vartheta,\varphi) = c_l e^{il\varphi}\sin^l(\vartheta), \tag{3.100}$$

with $c_l$ the renormalization constant

$$c_l = \left[\frac{(-1)^l}{2^l l!}\right]\sqrt{\frac{(2l+1)(2l)!}{4\pi}}. \tag{3.101}$$

From this we can obtain subsequent $Y_l^m(\vartheta,\varphi)$, which for $m \geq 0$ is

$$Y_l^m(\vartheta,\varphi) = \frac{(-1)^l}{2^l l!}\sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\varphi}\frac{1}{\sin^m\vartheta}\frac{d^{l-m}}{d(\cos\vartheta)^{l-m}}(\sin\vartheta)^{2l}, \tag{3.102}$$

and we define

$$Y_l^{-m} = (-1)^m[Y_l^m(\vartheta,\varphi)]^*. \tag{3.103}$$

Note here that we cannot have $l$ or $m$ be half-integer valued since this would give us a multivalued function since we acquire a minus sign under a $2\pi$ rotation.

### 3.6.3 Spherical Harmonics as Rotation Matrices

We can find a general direction eigenket by applying a rotation $\mathscr{D}(R)$ to $|\hat{\mathbf{z}}\rangle$ such that

$$|\hat{\mathbf{n}}\rangle = \mathscr{D}(R)|\hat{\mathbf{z}}\rangle. \tag{3.104}$$

We can find this by two rotation, first about the $y$-axis and then about the $z$-axis, which in the language of Euler rotations is

$$\mathscr{D}(R) = \mathscr{D}(\alpha = \varphi, \beta = \vartheta, \gamma = 0). \tag{3.105}$$

Inserting the completeness relation in the rotation we find

$$|\hat{\mathbf{n}}\rangle = \sum_l \sum_m \mathscr{R} = |l,m\rangle\langle l,m|\hat{\mathbf{z}}\rangle. \tag{3.106}$$

We find that $|\hat{\mathbf{n}}\rangle$ expanded in $|l,m\rangle$ contains all possible $l$. We now multiply by $\langle l,m'|$ on the left and find

$$\langle l,m'|\hat{\mathbf{n}}\rangle = \sum_m \mathscr{D}_{mm'}^{(l)}(\varphi,\vartheta,0)\langle l,m|\hat{\mathbf{z}}\rangle, \tag{3.107}$$

and we see that $\langle l,m|\hat{\mathbf{z}}\rangle = Y_l^(m*)(0,\varphi)$ where $\varphi$ is undetermined. Using this we find

$$Y_l^{m**}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\mathscr{D}_{m'0}^{(l)}(\varphi,\vartheta,0) \tag{3.108}$$

or

$$D_{m0}^{(l)}(\varphi,\vartheta,0) = \sqrt{\frac{4\pi}{2l+1}}Y_L^{m*}(\vartheta,\varphi) \tag{3.109}$$

## 3.8 Addition of Angular Momenta

### 3.8.1 Simple Examples of Angular-Momentum Addition

We have only studied either spin $\frac{1}{2}$ systems with other degrees of freedom ignored, or space systems with the internal degrees of freedom, such as spin ignored. Before going into the formal theory we will consider some familiar examples.

We consider the state that is a direct product state of the position and spin kets $|\mathbf{x}', \updownarrow\rangle = |\mathbf{x}'\rangle \otimes |\updownarrow\rangle$, where operators in the space spanned by the position kets commute by the operators in the space spanned by the spin-space. The rotation operator will still be $\exp(i\mathbf{J}\cdot\hat{\mathbf{n}}\varphi/\hbar)$, but angular momentum, the generator of rotation, will be

$$\mathbf{J} = \mathbf{L} + \mathbf{S} = \mathbf{L}\otimes 1 + 1\otimes\mathbf{S}. \tag{3.110}$$

Since $[\mathbf{L}, \mathbf{S}] = 0$ we can write

$$\mathscr{D}(R) = \mathscr{D}^{(\mathrm{orb})}\otimes\mathscr{D}^{(\mathrm{spin})} = \exp\left(-\frac{i\mathbf{L}\cdot\hat{\mathbf{n}}}{\hbar}\varphi\right)\otimes\exp\left(-\frac{i\mathbf{S}\cdot\hat{\mathbf{n}}}{\hbar}\varphi\right). \tag{3.111}$$

We can write the wave function for a spin particle as

$$\langle\mathbf{x}', \updownarrow|\alpha\rangle = \psi_{\updownarrow}(\mathbf{x}') \tag{3.112}$$

or as a column matrix

$$\begin{pmatrix}\psi_{\uparrow}(\mathbf{x}')\\ \psi_{\downarrow}(\mathbf{x}')\end{pmatrix}. \tag{3.113}$$

We might change the representation of the space part to be the eigenket $|n, l, m\rangle$.

If we instead consider a spin $\frac{1}{2}$ two particle system, without considering the orbital angular momentum we can write the total spin operator

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 = \mathbf{S}_1\otimes 1 + 1\otimes\mathbf{S}_2, \tag{3.114}$$

where it is evident that operators living in the two different spaces commute. The usual commutation relations for the operators in the same space, and for the total operator applies, and we can write the eigenvalues as

$$\mathbf{S}^2 = (\mathbf{S}_1 + \mathbf{S}_2)^2 : s(s+1)\hbar^2 \tag{3.115}$$

$$S_z = S_{1z} + S_{2z} : m\hbar \tag{3.116}$$

$$S_{1z} : m_1\hbar \tag{3.117}$$

$$S_{2z} : m_2\hbar. \tag{3.118}$$

We can expand the ket corresponding of an arbitrary spin of the two particle system by either eigenkets to $\mathbf{S}^2$ and $S_z$ or eigenkets to $S_{1z}$ and $S_{2z}$:

1. Expanding in the $\{m_1, m_2\}$ basis gives the possible kets

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, \text{and } |\downarrow\downarrow\rangle. \tag{3.119}$$

2. Expanding in the $\{s, m\}$ basis (also called the triplet-singlet representation) gives the possible kets ($|s, m\rangle$)

$$|0,0\rangle, |1,-1\rangle, |1,0\rangle, \text{and } |1,1\rangle. \tag{3.120}$$

The relationship between these two bases are

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \tag{3.121}$$
$$|1,-1\rangle = |\downarrow\downarrow\rangle \tag{3.122}$$
$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \tag{3.123}$$
$$|1,1\rangle = |\uparrow\uparrow\rangle. \tag{3.124}$$

The coefficients that appear on the right-hand side are the simplest version of the Clebsch-Gordan coefficients. These coefficients are the matrix elements of the matrix that transforms between two different bases.

### 3.8.2 Formal Theory of Angular-Momentum Addition

We now consider two angular-momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$, living in different subspaces. The individual components of each operator satisfies the normal commutation relations, but they commute with the components of the other operator. Looking at the infinitesimal rotation of both subspaces simultaneously, we see

$$\left(1 - i\frac{\mathbf{J}_1 \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi\right) \otimes \left(1 - i\frac{\mathbf{J}_2 \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi\right) = 1 - i\frac{(\mathbf{J}_1 \otimes 1 + 1 \otimes \mathbf{J}_2) \cdot \hat{\mathbf{n}}}{\hbar}\delta\varphi. \tag{3.125}$$

We define the total angular momentum operator as

$$\mathbf{J} = \mathbf{J}_1 \otimes 1 + 1 \otimes \mathbf{J}_2 = \mathbf{J}_1 + \mathbf{J}_2. \tag{3.126}$$

For a finite rotation we have

$$\mathscr{D}_1(R) \otimes \mathscr{D}_2(R) = \exp\left(-i\frac{\mathbf{J}_1 \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right) \otimes \exp\left(-i\frac{\mathbf{J}_2 \cdot \hat{\mathbf{n}}}{\hbar}\varphi\right). \tag{3.127}$$

Note the apparent similarities of the angle and axis of rotation for both subspaces. Importantly, the components of $\mathbf{J}$ satisfy the commutation relation for angular momentum. Physically the total angular momentum operator is the generator of rotation for the entire system. For choosing base kets there are two options:

1. The simultaneous eigenkets of $\mathbf{J}_1^2, \mathbf{J}_2^2, J_{1z}$, and $J_{2z}$, or the $\{j_1, j_2, m_1, m_2\}$ basis, denoted $|j_1, j_2; m_1, m_2\rangle$. The defining equations are

$$\mathbf{J}_1^2|j_1, j_2; m_1, m_2\rangle = j_1(j_1+1)\hbar^2|j_1, j_2; m_1, m_2\rangle \tag{3.128a}$$
$$\mathbf{J}_2^2|j_1, j_2; m_1, m_2\rangle = j_2(j_2+1)\hbar^2|j_1, j_2; m_1, m_2\rangle \tag{3.128b}$$
$$J_{1z}|j_1, j_2; m_1, m_2\rangle = m_1\hbar|j_1, j_2; m_1, m_2\rangle \tag{3.128c}$$
$$J_{2z}|j_1, j_2; m_1, m_2\rangle = m_2\hbar|j_1, j_2; m_1, m_2\rangle \tag{3.128d}$$

2. The simultaneous eigenkets of $\mathbf{J}_1^2, \mathbf{J}_2^2, \mathbf{J}^2$, and $J_z$ or the $\{j_1, j_2; j, m\}$ basis. We note that this set commutes, denoted $|j_1, j_2; j, m\rangle$. The defining equation are

$$\mathbf{J}_1^2|j_1, j_2; j, m\rangle = j_1(j_1+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129a}$$
$$\mathbf{J}_2^2|j_1, j_2; j, m\rangle = j_2(j_2+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129b}$$
$$\mathbf{J}^2|j_1, j_2; j, m\rangle = j(j+1)\hbar^2|j_1, j_2; j, m\rangle \tag{3.129c}$$
$$J_z|j_1, j_2; j, m\rangle = m\hbar|j_1, j_2; j, m\rangle. \tag{3.129d}$$

   Commonly the values for $j_1, j_2$ are implicit, and the ket is denoted as just $|j, m\rangle$.

It is important to note that

$$\left[\mathbf{J}^2, J_z\right] = 0, \quad \left[\mathbf{J}^2, J_{1z}\right] \neq 0, \quad \left[\mathbf{J}^2, J_{2z}\right] \neq 0. \tag{3.130}$$

We can therefore not mix these two representations. Consider the connecting transformation for the two bases

$$|j_1, j_2; j, m\rangle = \sum_{m_1}\sum_{m_2} |j_1, j_2; m_1, m_2\rangle\langle j_1, j_2; m_1, m_2|j_1, j_2; j, m\rangle. \tag{3.131}$$

The elements $\langle j_1, j_2; m_1, m_2|j_1, j_2; j, m\rangle$ are the Clebsch-Gordan coefficients.

We will now study some properties of the Clebsch-Gordan coefficients. If

$$m \neq m_1 + m_2, \tag{3.132}$$

the coefficients vanish. The coefficients also vanish unless

$$|j_1 - j_2| \leq j \leq j_1 + j_2. \tag{3.133}$$

The Clebsch-Gordan coefficients also form a unitary matrix, and we, by convention, take the elements to be real. A consequence of this (being a real unitary matrix) is that the matrix is orthogonal. We can write the Clebsch-Gordan coefficients in Wigner's 3-j symbol

$$\langle j_1, j_2; m_1, m_2|j_1, j_2; j, m\rangle = (-1)^{j_1 - j_2 + m}\sqrt{2j+1}\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix}. \tag{3.134}$$

#### 3.8.3 Recursion Relations for the Clebsch-Gordan Coefficients

#### 3.8.4 Clebsch-Gordan Coefficients and Rotation Matrices

### 3.11 Tensor Operators

#### 3.11.1 Vector Operator

#### 3.11.2 Cartesian Tensors Versus Irreducible Tensors

#### 3.11.3 Product of Tensors

#### 3.11.4 Matrix Elements of Tensor Operators; The Wigner-Eckart Theorem

## 4 Symmetry in Quantum Mechanics

### 4.1 Symmetries, Conservation Laws, and Degeneracies

#### 4.1.1 Symmetries in Classical Physics

We will review the concept of symmetry and conservation in classical physics. Suppose we have a Lagrangian $L$, a function of the generalized coordinate $q_i$ and the corresponding velocity $\dot{q}_i$. For a displacement

$$q_i \to q_i + \delta q_i, \tag{4.1}$$

if the Lagrangian remains unchanged, we have

$$\frac{\partial L}{\partial q_i} = 0. \tag{4.2}$$

From the Lagrange equation it then follows that

$$\frac{dp_i}{dt} = 0, \tag{4.3}$$

where the canonical momentum is defined by

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{4.4}$$

So the canonical momentum conjugate to $q_i$ is a conserved quantity under displacement if $L$ remains unchanged.

Similarly, looking at the Hamiltonian formulation we have $H$ as a function of $q_i$ and $p_i$, we have

$$\frac{dp_i}{dt} = 0 \tag{4.5}$$

when

$$\frac{\partial H}{\partial t} = 0. \tag{4.6}$$

Thus, if we lack an explicit $q_i$ dependence in $H$, ($H$ has a symmetry under $q_i \to q_i + \delta q_i$) we have a conserved quantity.

### 4.1.2 Symmetry in Quantum Mechanics

In quantum mechanics, we have made the association between an operation, like translation or rotation, to a unitary operator, like $\mathscr{S}$. Regardless if the physical system has a symmetry corresponding to the operator $\mathscr{S}$, we tend to call this operator for a symmetry operator. Furthermore, we have seen that if the symmetry operation is infinitesimal we can write

$$\mathscr{S} = 1 - \frac{i\varepsilon}{\hbar} G, \tag{4.7}$$

where $G$ is the Hermitian generator of the symmetry operator.

Consider that $H$ is invariant under $\mathscr{S}$. From this, it follows that

$$\mathscr{S}^\dagger H \mathscr{S} = H, \tag{4.8}$$

or equivalently

$$[G, H] = 0. \tag{4.9}$$

From Heisenberg's equation of motion ($i\hbar\partial_t A = [A, H]$), we obtain

$$\frac{dG}{dt} = 0, \tag{4.10}$$

and thus, $G$ is a constant of the motion. E.g. if the Hamiltonian is translationally invariant, the momentum (that generates the translation), is a constant of the motion.

Consider the system in an eigenket of $G$, when $G$ commutes with $H$, at time $t_0$. For a later time $t > t_0$, we have the ket

$$|g', t_0; t\rangle = U(t, t_0)|g'\rangle, \tag{4.11}$$

which is also an eigenket of $G$ the same eigenvalue $g'$. Thus, when a ket is an eigenket to $G$ with eigenvalue $g'$ it will stay as such. Proving this is just realizing that since $[G, H] = 0$, then $[G, U] = 0$.

### 4.1.3 Degeneracies

Suppose that for a symmetry operator $\mathscr{S}$ we have

$$[H, \mathscr{S}] = 0, \tag{4.12}$$

and $|n\rangle$ is an energy eigenket with eigenvalue $E_n$. Evidently, $\mathscr{S}|n\rangle$ is also an energy eigenket with the same energy. If $|n\rangle$ and $\mathscr{S}|n\rangle$ represent different states we say that the states are degenerate. Often, the symmetry operator is a function of a continuous parameter, say $\lambda$, and thus all states of the form $\mathscr{S}(\lambda)|n\rangle$, are degenerate.

Consider that the symmetry operator in question is rotation, that is we have $\mathscr{D}(R)$. We also suppose that the Hamiltonian is rotationally invariant

$$[\mathscr{D}(R), H] = 0. \tag{4.13}$$

This implies the commutation relations

$$[\mathbf{J}, H] = 0 \tag{4.14a}$$

$$\left[\mathbf{J}^2, H\right] = 0, \tag{4.14b}$$

and we can have simultaneous eigenkets of $H$, $\mathbf{J}^2$, and $J_z$, denoted by $|n; j, m\rangle$. All states on the form

$$\mathscr{R}|n; j, m\rangle \tag{4.15}$$

are degenerate and share the same energy. During rotation, different $m$ values are mixed, such that generally, $\mathscr{D}(R)|n; j, m\rangle$ is a linear combination of $2j + 1$ independent states. We can write this as

$$\mathscr{D}(R)|n; j, m\rangle = \sum_{m'} |n; j, m'\rangle \mathscr{D}^{(j)}_{m'm}(R). \tag{4.16}$$

This degeneracy is equal to the number of $m$-values, that is $2j + 1$. It is also possible to reach this conclusion by applying $J_\pm$ to the ket.

As an example of this consider an atomic electron subject to the potential $V = V(r) + V_{LS}(r)\mathbf{L} \cdot \mathbf{S}$. Since this potential is rotationally invariant, we expect a $(2j + 1)$-fold degeneracy. If we applied an external magnetic field, this symmetry would break, and we would get distinct energy levels.

### 4.1.4 SO(4) Symmetry in the Coulomb Potential

An archetypal example of continuous symmetry in quantum mechanics is the solution of the coulomb potential in the hydrogen atom. The energy eigenvalue for this potential is

$$E_n = \frac{1}{2} mc^2 \frac{Z^2\alpha^2}{n^2} = -13.6\,\mathrm{eV} \frac{Z^2}{n^2}, \tag{4.17}$$

and since this energy is only dependent on $n$, the degeneracy for the ket $|n, l, m\rangle$ is

$$\sum_{l=0}^{n-1} (2l + 1) = n^2. \tag{4.18}$$

This is due to the symmetry of $1/r$ potentials.

Classically, the orbits of $1/r$ potentials have been studied, namely the Kepler problem. The result from this leads us to the conclusion of the existence of a constant of the motion, which in classical mechanics is

$$\mathbf{M} = \frac{\mathbf{p} \times \mathbf{L}}{m} - \frac{Ze^2}{r}\mathbf{x}, \tag{4.19}$$

called the Lenz vector, or the Runge-Lenz vector.

Treating this quantum mechanically, this new symmetry is called SO(4), analogous to the SO(3) group, but instead rotation operators in four spatial dimensions. We start by modifying the Lenz vector such that it becomes Hermitian

$$\mathbf{M} = \frac{1}{2m}(\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p}) - \frac{Ze^2}{r}\mathbf{x}. \tag{4.20}$$

We then have

$$[\mathbf{M}, H] = 0, \tag{4.21}$$

for the Hamiltonian

$$H = \frac{\mathbf{p}^2}{2m} - \frac{Ze^2}{r}, \tag{4.22}$$

and thus $\mathbf{M}$ is a constant of the motion. Furthermore, the relations

$$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = 0, \tag{4.23}$$

$$\mathbf{M}^2 = \frac{2}{m}H(\mathbf{L}^2 + \hbar^2) + Z^2e^4, \tag{4.24}$$

are useful.

We want to identify the responsible symmetry for this constant of the motion. This is done by exploring the algebra of the generators of this symmetry. We know

$$[L_i, L_j] = i\hbar\varepsilon_{ijk}L_k, \tag{4.25}$$

and it is possible to show

$$[M_i, L_j] = i\hbar\varepsilon_{ijk}M_k. \tag{4.26}$$

With this we can derive that

$$[M_i, M_j] = -i\hbar\varepsilon_{ijk}\frac{2}{m}HL_k. \tag{4.27}$$

These commutation relations do not form a closed algebra as a consequence of the presence of $H$, making it challenging identifying these operators as the generators. Conversely, we will consider the problem with specific bound states. The vector space will now only contain the eigenstates to $H$ that have an energy eigenvalue of $E < 0$, and we may replace $H$ with $E$, closing the algebra. We introduce a scaled vector operator

$$\mathbf{N} = \sqrt{-\frac{m}{2E}}\mathbf{M}, \tag{4.28}$$

such that the closed algebra reads as

$$[L_i, L_j] = i\hbar\varepsilon_{ijk}L_k \tag{4.29a}$$

$$[N_i, L_j] = i\hbar\varepsilon_{ijk}N_k \tag{4.29b}$$

$$[N_i, N_j] = i\hbar\varepsilon_{ijk}L_k, \tag{4.29c}$$

The question becomes what type of symmetry is generated by the operators $\mathbf{L}$ and $\mathbf{N}$? Interestingly, the answer is rotation in four spatial dimensions. The first clue for this is that the required number of operators for rotation in $n$ dimensions is given by $n(n-1)/2$, and thus rotation in four dimensions require six operators, which we have with $\mathbf{L}$ and $\mathbf{n}$. It is less obvious to see that the above algebra is appropriate for this.

We generalize the procedure performed for three dimensions with $\mathbf{L}$. We start by rewriting $(x, y, z)$ and $(p_x, p_y, p_z)$ to $(x_1, x_2, x_3)$ and $(p_1, p_2, p_3)$. Consequently, we write

$$L_1 = \tilde{L}_{23} = x_2 p_3 - x_3 p_2, \tag{4.30a}$$
$$L_2 = \tilde{L}_{31} = x_3 p_1 - x_1 p_3, \tag{4.30b}$$
$$L_3 = \tilde{L}_{12} = x_1 p_2 - x_2 p_1. \tag{4.30c}$$

We know invent a fourth spatial dimension $x_4$ and define its conjugate momentum $p_4$ according to the canonical commutation relation, and write

$$N_1 = \tilde{L}_{14} = x_1 p_4 - x_4 p_1, \tag{4.31a}$$
$$N_2 = \tilde{L}_{24} = x_2 p_4 - x_4 p_2, \tag{4.31b}$$
$$N_3 = \tilde{L}_{34} = x_3 p_4 - x_4 p_3. \tag{4.31c}$$

These operators obey the closed algebra defined earlier. That is, the closed algebra we found, is the algebra for rotation in four spatial dimensions.

We continue by investigating the degeneracies in the Coulomb potential. Starting by defining the operators

$$\mathbf{I} = \frac{\mathbf{L} + \mathbf{N}}{2}, \tag{4.32}$$
$$\mathbf{K} = \frac{\mathbf{L} - \mathbf{N}}{2}, \tag{4.33}$$

we can prove the algebra

$$[I_i, I_j] = i\hbar\varepsilon_{ijk} I_k \tag{4.34a}$$
$$[K_i, K_j] = i\hbar\varepsilon_{ijk} K_k \tag{4.34b}$$
$$[I_i, K_j] = 0. \tag{4.34c}$$

As such, these operators obey independent angular momentum algebra, and we also have that $[\mathbf{I}, I] = [\mathbf{K}, H] = 0$. We denote the eigenvalues of $\mathbf{I}^2$ and $\mathbf{K}^2$ by $i(i+1)\hbar^2$ and $k(k+1)\hbar^2$ respectively, with $i, k$ taking on positive half-integer values.

We find that

$$\mathbf{I}^2 - \mathbf{K}^2 = \frac{1}{4}\left[(\mathbf{L} + \mathbf{N})^2 - (\mathbf{L} - \mathbf{N})^2\right] \tag{4.35a}$$
$$= \frac{1}{4}\left[(\mathbf{L}^2 + \mathbf{N}^2 + 2\mathbf{L}\cdot\mathbf{N}) - (\mathbf{L}^2 + \mathbf{N}^2 - 2\mathbf{L}\cdot\mathbf{N})\right] \tag{4.35b}$$
$$= \mathbf{L}\cdot\mathbf{N} \tag{4.35c}$$
$$= 0, \tag{4.35d}$$

and thus we must have that $i = k$. Similarly, we find

$$\mathbf{I}^2 + \mathbf{K}^2 = \frac{1}{4}\big[(\mathbf{L}+\mathbf{N})^2 + (\mathbf{L}-\mathbf{N})^2\big] \tag{4.36a}$$
$$= \frac{1}{4}\big[(\mathbf{L}^2+\mathbf{N}^2+2\mathbf{L}\cdot\mathbf{N}) + (\mathbf{L}^2+\mathbf{N}^2-2\mathbf{L}\cdot\mathbf{N})\big] \tag{4.36b}$$
$$= \frac{1}{2}\big[\mathbf{L}^2+\mathbf{N}^2\big] \tag{4.36c}$$
$$= \frac{1}{2}\Big[\mathbf{L}^2 - \frac{m}{2E}\mathbf{M}^2\Big], \tag{4.36d}$$

that together with Eq. (4.24) gives

$$2k(k+1)\hbar^2 = \frac{1}{2}\Big(-\hbar^2 - \frac{m}{2E}Z^2e^4\Big). \tag{4.37}$$

Solving this for $E$ yields the relation

$$E = -\frac{mZ^2e^4}{2\hbar^2}\frac{1}{(2k+1)^2}. \tag{4.38}$$

This is the energy for the hydrogen atom but with $n$ replaced with $2k+1$. Thus, the rotational symmetries that arise from $\mathbf{I}$ and $\mathbf{K}$ creates the degeneracy seen. Technically the degeneracy is $(2i+1)(2k+1) = (2k+1)^2 = n^2$.

Note here that we found the eigenvalues of the hydrogen atom without solving the Schrödinger equation. Instead, we used the symmetries in the system to obtain the same answer.

The SO(4) algebra used here can also, as can be seen from the above calculations be thought of as SU(2)×SU(2), as $\mathbf{I}$ and $\mathbf{K}$ independently follow the SU(2) group algebra.

## 4.2 Discrete Symmetries, Parity, or Space Inversion

Until now, we have considered symmetry operators that are continuous, that is they can be obtained by applying infinitesimal operations successively. We will now consider three types of discrete symmetry operations: Parity, lattice translation, and time reversal.

Consider parity (space inversion). The parity operation, when applied to a coordinate system changes it from a right-handed system to a left-handed system. We will however, consider the parity operator applied to the state kets. Take a general state ket $|\alpha\rangle$ and apply the parity operator $\pi$ to obtain a space inverted state

$$|\alpha\rangle \to \pi|\alpha\rangle. \tag{4.39}$$

We require from the parity operator that the expectation value of $\mathbf{x}$ must have the opposite sign when taken with a space-inverted state:

$$\big\langle\alpha\big|\pi^\dagger\mathbf{x}\pi\big|\alpha\big\rangle = -\langle\alpha|\mathbf{x}|\alpha\rangle, \tag{4.40}$$

which is to say that we have

$$\pi^\dagger\mathbf{x}\pi = -\mathbf{x}, \tag{4.41}$$

or

$$\mathbf{x}\pi = -\pi\mathbf{x}. \tag{4.42}$$

We know that $\pi$ is unitary, and we can see that $\pi$ and $\mathbf{x}$ must anticommute.

Noting that for an eigenket to the position operator, we have

$$\mathbf{x}\pi|\mathbf{x}'\rangle = -\pi\mathbf{x}|\mathbf{x}'\rangle = (-\mathbf{x}')\pi|\mathbf{x}'\rangle. \tag{4.43}$$

Thus, $\pi|\mathbf{x}'\rangle$ is still an eigenket of $\mathbf{x}$, but with the eigenvalue $-\mathbf{x}'$, up to a global phase factor, which we generally take to be 1, that is $e^{i\delta} = 1$ for $\delta$ real. With this we see that

$$\pi^2|\mathbf{x}'\rangle = |\mathbf{x}'\rangle, \tag{4.44}$$

and thus $\pi^2 = 1$, and $\pi$ is therefore Hermitian:

$$\pi^{-1} = \pi^\dagger = \pi, \tag{4.45}$$

with eigenvalue either $+1$ or $-1$.

Consider now the momentum operator $\mathbf{p}$ as the generator of translation. A translation followed by parity, is equivalent to parity followed by a translation in the reverse direction. Thus,

$$\pi\mathscr{J}(d\mathbf{x}') = \mathscr{J}(-d\mathbf{x}')\pi, \tag{4.46}$$

or equivalently

$$\pi\left(1 - \frac{i\mathbf{p}\cdot d\mathbf{x}'}{\hbar}\right)\pi^\dagger = 1 + \frac{i\mathbf{p}\cdot d\mathbf{x}'}{\hbar}. \tag{4.47}$$

From the above we conclude

$$\{\pi, \mathbf{p}\} = 0 \tag{4.48}$$

and

$$\pi^\dagger\mathbf{p}\pi = -\mathbf{p}. \tag{4.49}$$

Now take the angular momentum operator $\mathbf{J}$ as a generator of rotation. For $3\times 3$ orthogonal matrices, we have

$$R^{(\text{parity})}R^{(\text{rotation})} = R^{(\text{rotation})}R^{(\text{parity})}, \tag{4.50}$$

for the explicit parity matrix

$$R^{(\text{parity})} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{4.51}$$

Thus, parity and rotation operations commute. We then postulate the corresponding relation for operators

$$\pi\mathscr{D}(R) = \mathscr{D}(R)\pi. \tag{4.52}$$

We can now see that

$$[\pi, \mathbf{J}] = 0 \tag{4.53}$$

and

$$\pi^\dagger\mathbf{J}\pi = \mathbf{J}. \tag{4.54}$$

This holds for $\mathbf{L}$ and $\mathbf{S}$ individually as well.

Under rotations, $\mathbf{x}$ and $\mathbf{J}$ transform as vectors, or spherical tensors of rank 1. However, $\mathbf{x}$ and $\mathbf{p}$ are both odd under parity while $\mathbf{L}$ is even under parity. Odd parity vectors are called polar vectors, while even parity vectors are called axial vectors, or pseudovectors.

Suppose we have an operation on the form $\mathbf{S}\cdot\mathbf{x}$, that is a dot product between a polar and axial vector. Under rotation, they behave as a normal scalar (like $\mathbf{S}\cdot\mathbf{L}$ or $\mathbf{x}\cdot\mathbf{p}$), under parity however, they behave like

$$\pi^{\dagger}\mathbf{S}\cdot\mathbf{x} = -\mathbf{S}\cdot\mathbf{x}, \tag{4.55}$$

while for normal scalars we have

$$\pi^{\dagger}\mathbf{L}\cdot\mathbf{S}\pi = \mathbf{L}\cdot\mathbf{S}. \tag{4.56}$$

We call the operator $\mathbf{S}\cdot\mathbf{x}$ for a pseudoscalar.

### 4.2.1 Wave Functions and Parity

We consider a spinless particle with the wave function

$$\psi(\mathbf{x}') = \langle\mathbf{x}'|\alpha\rangle. \tag{4.57}$$

For a space inverted state, that is a state with the ket $\pi|\alpha\rangle$, has the wave function

$$\langle\mathbf{x}'|\pi|\alpha\rangle = \langle-\mathbf{x}'|\alpha\rangle = \psi(-\mathbf{x}'). \tag{4.58}$$

If we take $|\alpha\rangle$ to be an eigenket of parity, we have

$$\pi|\alpha\rangle = \pm|\alpha\rangle. \tag{4.59}$$

The wave function is then

$$\langle\mathbf{x}'|\pi|\alpha\rangle = \pm\langle\mathbf{x}'|\alpha\rangle. \tag{4.60}$$

This combined with the earlier wave function gives, depending on the parity of $|\alpha\rangle$,

$$\psi(-\mathbf{x}') = \pm\psi(\mathbf{x}') \begin{cases} + & \text{even parity} \\ - & \text{odd parity.} \end{cases} \tag{4.61}$$

Note that not all wave functions that are of interest for physical systems follow a parity law like the one above. For example, since the momentum operator anticommutes with the parity operator, the momentum eigenket is not a parity eigenket. We can see that the plane wave function does not obey the parity relation above.

Eigenkets of the orbital angular momentum however, are expected to be parity eigenkets since $[\pi,\mathbf{L}] = 0$. We investigate this by considering the wave function under space inversion

$$\langle\mathbf{x}'|\alpha;l,m\rangle = R_{\alpha}(r)Y_l^m(\vartheta,\varphi). \tag{4.62}$$

The parity transformation $\mathbf{x}' \to -\mathbf{x}'$ is accomplished for spherical coordinates by

$$r \to r \tag{4.63a}$$

$$\vartheta \to \pi - \vartheta \quad (\cos\vartheta \to -\cos\vartheta) \tag{4.63b}$$

$$\varphi \to \varphi + \pi \quad (e^{im\varphi} \to (-1)^m e^{im\varphi}). \tag{4.63c}$$

Using this and the definition of the spherical harmonics we can show that

$$Y_l^m = (-1)^l Y_l^m. \tag{4.64}$$

Therefore, we get the transformation under parity to be

$$\pi|\alpha;l,m\rangle = (-1)^l|\alpha;l,m\rangle. \tag{4.65}$$

We will now consider the parity properties of the energy eigenkets. A very important theorem reads:

**Theorem** Suppose

$$[H, \pi] = 0 \tag{4.66}$$

and $|n\rangle$ is a nondegenerate eigenket of $H$ with eigenvalue $E_n$:

$$H|n\rangle = E_n|n\rangle; \tag{4.67}$$

then $|n\rangle$ is also a parity eigenket.

**Proof** First consider the state

$$|\alpha\rangle = \frac{1}{2}(1 \pm \pi)|n\rangle. \tag{4.68}$$

Then we show

$$\pi|\alpha\rangle = \frac{1}{2}(\pi \pm \pi^2)|n\rangle = \frac{1}{2}(\pi \pm 1)|n\rangle = \pm|\alpha\rangle, \tag{4.69}$$

and thus $\alpha$ is a parity eigenket. Since $H$ and $\pi$ commute it is easy to also show that $|\alpha\rangle$ is also an energy eigenket with eigenvalue $E_n$. Thus, $|n\rangle$ and $|\alpha\rangle$ must be the same up to a multiplicative constant, and $|n\rangle$ must therefore be a parity eigenket with eigenvalue $\pm 1$. Note also that despite the $\pm$, $|\alpha\rangle$ defines just one state. □

We now look at the harmonic oscillator as an example. The ground state $|0\rangle$, has a Gaussian wave function, which is even under $\mathbf{x}' \to -\mathbf{x}'$, and thus $|0\rangle$ has even parity. The first excited state

$$|1\rangle = a^\dagger|0\rangle, \tag{4.70}$$

must have odd parity since $a^\dagger$ is linear in $x$ and $p$, both of which are odd. Continuing this line of reasoning we can show that the $n$th state of the harmonic operator $|n\rangle$ has the parity $(-1)^n$. Important to note here is that this result relies on the assumption of non-degeneracy.

Another example is the momentum eigenket. Since momentum anticommutes with parity, the momentum eigenket is not a parity eigenket, even though the free particle Hamiltonian is invariant under parity. The reason being that we have a degeneracy with $|\mathbf{p}'\rangle$ and $|-\mathbf{p}'\rangle$ having the same energy. Interestingly we can construct eigenkets

$$\frac{1}{\sqrt{2}}(|\mathbf{p}\rangle' \pm |-\mathbf{p}'\rangle) \tag{4.71}$$

which are parity eigenkets. This is in wave functions, equivalent to saying that the plane wave function $e^{i\mathbf{p}'\cdot\mathbf{x}'/\hbar}$ does not have definite parity, while the wave functions $\cos(\mathbf{p}'\cdot\mathbf{x}'/\hbar)$ and $\sin(\mathbf{p}'\cdot\mathbf{x}'/\hbar)$ do.

### 4.2.2 Symmetrical Double-Well Potential

We consider a symmetrical double-well potential. The Hamiltonian is invariant under parity and the two lowest solutions we call the symmetrical state $|S\rangle$ and the antisymmetrical state $|A\rangle$. These two states are simultaneous eigenkets of both $H$ and $\pi$, and we have

$$E_A > E_S. \tag{4.72}$$

We form the states

$$|R\rangle = \frac{1}{\sqrt{2}}(|S\rangle + |A\rangle) \tag{4.73a}$$

$$|L\rangle = \frac{1}{\sqrt{2}}(|S\rangle - |A\rangle). \tag{4.73b}$$

These two states are neither parity states, being interchanged under parity, or energy eigenstates. They are examples of nonstationary states. Consider that at $t = 0$ we have the state $|R\rangle$, then at a later time we have

$$|R, t_0 = 0; t\rangle = U(t, 0)|R, 0; 0\rangle \tag{4.74a}$$

$$= \frac{1}{\sqrt{2}}(e^{-iE_S t/\hbar}|S\rangle + e^{iE_A t/\hbar}|A\rangle) \tag{4.74b}$$

$$= \frac{e^{-iE_S t/\hbar}}{\sqrt{2}}(|S\rangle + e^{i(E_A - E_S)t/\hbar}|A\rangle. \tag{4.74c}$$

At a time $t = T/2 = \pi\hbar/(E_A - E_S)$ we find the system in $|S\rangle$. At the time $t = T$, we have $|R\rangle$ again, and we continue oscillating between these two states with angular frequency

$$\omega = \frac{E_A - E_S}{\hbar}. \tag{4.75}$$

This can be seen as tunnelling between the left and right states. If we increase the middle barrier to infinity, the states $|S\rangle$ and $|A\rangle$ become degenerate. Thus, $|L\rangle$ and $|R\rangle$ are now energy eigenkets, but they are not parity eigenkets. The infinitely high barrier makes tunnelling impossible. Interestingly, since we have degeneracy, the ground state can be asymmetrical, even though the Hamiltonian is symmetrical under parity.

### 4.2.3 Parity-Selection Rule

We consider the parity eigenstates $|\alpha\rangle$ and $|\beta\rangle$, such that

$$\pi|\alpha\rangle = \varepsilon_\alpha|\alpha\rangle \tag{4.76a}$$

$$\pi|\beta\rangle = \varepsilon_\beta|\beta\rangle, \tag{4.76b}$$

where $\varepsilon_\alpha, \varepsilon_\beta$ are the parity eigenvalues $\pm 1$. Unless $\varepsilon_\alpha = -\varepsilon_\beta$, we have

$$\langle\beta|\mathbf{x}|\alpha\rangle = 0. \tag{4.77}$$

Thus, the odd-parity operator $\mathbf{x}$ connects parity states of different parity. There is a similar argument due to Wigner

$$\int d\tau \psi_\beta^* \mathbf{x} \psi_\alpha = 0, \tag{4.78}$$

if $\psi_b$ and $\psi_a$ have the same parity. This rule was known before the birth of quantum mechanics, from analysis of spectral lines, and called Laporte's rule. Wigner then showed that this rule was due to parity selection. From Wigner's rule it follows that if the Hamiltonian is invariant under parity, then nondegenerate energy eigenstates cannot possess a permanent electric dipole moment, since they are also parity eigenstates.

Generally, operators that have odd parity, have nonvanishing matrix elements only between states of opposite parity, that is, they connect opposite parity states. For operators with even parity, the reverse is true, they connect same parity states.

### 4.2.4 Parity Nonconservation

The Hamiltonian for the weak interaction is not invariant under parity, and as such parity is not conserved through the weak interaction. We can therefore have decays with final states being superpositions of opposite parity states, and the observables can depend on pseudoscalars.

The first experiment that discovered this had a decay rate depending on $\langle \mathbf{S} \rangle \cdot \mathbf{P}$. The decay in question was ${}^{60}\mathrm{Co} \to {}^{60}\mathrm{Ni} + e^- + \bar{\nu}_e$, where the observables are the $\mathbf{S}$ for the cobalt nucleus and $\mathbf{p}$ is the momentum of the emitted electron.

Since the parity is not conserved in the weak interaction, what was once thought of as pure nuclear and atomic states, are parity mixtures.

## 4.3 Lattice Translation as a Discrete Symmetry

We will consider the discrete symmetry operation that is lattice transformation, with important applications in solid state physics. Consider a periodic potential in one dimension such that $V(x \pm a) = V(x)$. This can be thought of as a chain of positive ions, and an electron moving through this lattice. The Hamiltonian is not generally invariant under a translation $\tau(l)$ with $l$ arbitrary, where $\tau(l)$ has the properties

$$\tau^\dagger(l) x \tau(l) = x + l \tag{4.79a}$$
$$\tau(l)|x'\rangle = |x' + l\rangle. \tag{4.79b}$$

However, when $l = a$, with $a$ the lattice spacing we have

$$\tau^\dagger(a) V(x) \tau(a) = V(x + a) = V(x). \tag{4.80}$$

The kinetic energy part of the Hamiltonian is, of course, invariant under any type of displacement, so we have

$$\tau^\dagger(a) H \tau(a) = H, \tag{4.81}$$

and also $\tau(a)$ is unitary, with

$$[H, \tau(a)] = 0, \tag{4.82}$$

such that both $H$ and $\tau(a)$ can be diagonalized simultaneously. Note that $\tau(a)$ is not Hermitian, and we expect the eigenvalue to be complex modulus 1.

We will start by considering a periodic potential with the potential between two sites going to infinity. A state completely localized in a lattice site is a good ground state candidate. Assuming that the particle is localized at the $n$th site, denoted by $|n\rangle$ with the ground state energy $E_0$ such that $H|n\rangle = E_0|n\rangle$. The wave function $\langle x'|n\rangle$ is finite only on the $n$th site, but we note that similar states exists on any state $n$ where $n \in (-\infty, \infty)$.

The ket $|n\rangle$ is obviously not an eigenket to $\tau(a)$ since

$$\tau(a)|n\rangle = |n + 1\rangle. \tag{4.83}$$

We see once again that even though $\tau(a)$ and $H$ commute we do not have a simultaneous energy eigenket, consistent with us having an infinite-fold degeneracy. We want to find some ket that is a simultaneous eigenket of both $H$ and $\tau(a)$.

A similar procedure as performed earlier with the states $|L\rangle$ and $|R\rangle$ will be done here, where even though they were not parity eigenkets, some linear combination were. The linear combination

$$|\vartheta\rangle = \sum_{n=-\infty}^{\infty} e^{in\vartheta}|n\rangle, \tag{4.84}$$

with $\vartheta \in (-\pi, \pi]$ is formed. This is obviously an eigenket of $H$, since $|n\rangle$ is, and applying the lattice translation operator

$$\tau(a)|\vartheta\rangle = \sum_{-\infty}^{\infty} e^{in\vartheta}|n+1\rangle = \sum_{n=-\infty}^{\infty} e^{i(n-1)\vartheta}|n\rangle = e^{-i\vartheta}|\vartheta\rangle. \tag{4.85}$$

Thus, $|\vartheta\rangle$ is a simultaneous eigenket of $H$ and $\tau(a)$.

Returning to the lattice potential with finite height, albeit still tall enough. It is still possible to construct the kets $|n\rangle$ with property $\tau(a)|n\rangle = |n+1\rangle$, but the wave function $\langle x'|n\rangle$, can now leak into neighbouring sites due to quantum tunnelling effects. Assuming that the potential is high enough, we can ignore the leakage into sites other than the nearest neighbours. That is, we get the Hamiltonian

$$\langle n|H|n\rangle = E_0 \quad \text{and} \quad \langle n'|H|n\rangle = -\Delta \quad \text{only if} \quad n' = \pm 1. \tag{4.86}$$

This is called the tight-binding approximation in solid state physics. We thus get

$$H|n\rangle = E_0|n\rangle - \Delta|n+1\rangle - \Delta|n-1\rangle. \tag{4.87}$$

Note that $|n\rangle$ is no longer an energy eigenket.

We form that same linear combination as before

$$|\vartheta\rangle = \sum_{n=-\infty}^{\infty} e^{in\vartheta}|n\rangle. \tag{4.88}$$

This is still an eigenket to $\tau(a)$ since the steps performed earlier still holds. However, since $|n\rangle$ is not an energy eigenket any more, it is not obvious if $|\vartheta\rangle$ is. We test this by applying $H$

$$\begin{aligned}
H|\vartheta\rangle &= H\sum_n e^{in\vartheta}|n\rangle && \text{(4.89a)}\\
&= \sum_n e^{in\vartheta} H|n\rangle && \text{(4.89b)}\\
&= \sum_n e^{in\vartheta}(E_0|n\rangle - \Delta|n+1\rangle - \Delta|n-1\rangle) && \text{(4.89c)}\\
&= E_0\sum_n e^{in\vartheta}|n\rangle - \Delta\sum_n e^{in\vartheta}\left(e^{-i\vartheta} + e^{i\vartheta}\right)|n\rangle && \text{(4.89d)}\\
&= (E_0 - 2\Delta\cos\vartheta)\sum_n e^{in\vartheta}|n\rangle && \text{(4.89e)}\\
&= (E_0 - 2\Delta\cos\vartheta)|\vartheta\rangle. && \text{(4.89f)}
\end{aligned}$$

What is interesting here is that the eigenvalue depends on the parameter $\vartheta$ which it did not do for the infinite potential case. As $\Delta$ becomes non-zero the degeneracy is removed, and we get a continuous spectrum of energies in the range $E \in [E_0 - 2\Delta, E_0 + 2\Delta]$.

Now, we want to apply some physical significance to the parameter $\vartheta$, and we will therefore study the wave function $\langle x'|\vartheta\rangle$. The wave function for the lattice translated state $\tau(a)|\vartheta\rangle$ is

$$\langle x'|\tau(a)|\vartheta\rangle = \langle x' - a|\vartheta\rangle \tag{4.90}$$

or

$$\langle x'|\tau(a)|\vartheta\rangle = e^{-i\vartheta}\langle x'|\vartheta\rangle, \tag{4.91}$$

depending on if we let the operator act on the bra or ket. Thus,

$$\langle x' - a|\vartheta\rangle = \langle x'|\vartheta\rangle e^{-i\vartheta}, \tag{4.92}$$

which can be solved by setting

$$\langle x'|\vartheta\rangle = e^{ikx'} u_k(x'), \tag{4.93}$$

with $\vartheta = ka$. The function $u_k(x')$ is a periodic function with period $a$. With this we arrive at Bloch's theorem: The wave function of $|\vartheta\rangle$, which is an eigenket of $\tau(a)$, can be written as a plane wave $e^{ikx'}$ times a periodic function with period $a$. This theorem holds even if the tight-binding approximation breaks down.

We can now interpret the meaning of $|\vartheta\rangle$ and its energy eigenvalue. We know that the wave function is a plane wave characterized by the wave vector $k$ and modulated by the periodic function $u_k(x')$. As $\vartheta = ka$ varies between $-\pi$ and $\pi$ we have that the wave vector $k$ varies between $-\pi/a$ and $\pi/a$. The energy eigenvalue has a $k$ dependence

$$E(k) = E_0 - 2\Delta \cos ka. \tag{4.94}$$

This is independent of the shape of the potential as long as the tight-binding approximation holds. Note that we have a cutoff for the wave vector at $|k| = \pi/a$. The dispersion curve defined by $E(k)$ forms a continuous energy band between $E_0-2\Delta$ and $E_0+2\Delta$ called the Brillouin zone. Note also that we have lifted the infinite-fold degeneracy completely.

When considering multiple electrons, following the Pauli principle, as opposed to just a single electron as considered here, we can find many of the features of metals, semiconductors and insulators.

## 4.4 The Time-Reversal Discrete Symmetry

We will now explore the discrete symmetry operator that is time reversal. The term time reversal is a misnomer, what we mean are more accurately described by the term reversal of motion. This is the term used by Wigner when he formulated time reversal.

We start from a classical point of view. Imagine the trajectory of a particle in a certain force field. Suppose now that at $t = 0$ the particle stop and reverse its motion, that is $\mathbf{p}|_{t=0} \to -\mathbf{p}|_[t = 0]$. The particle thus moves backwards along the same path. We can say, more formally, that if $\mathbf{x}(t)$ is a solution to

$$m\ddot{\mathbf{x}}(t) = -\nabla V(\mathbf{x}), \tag{4.95}$$

then $\mathbf{x}(-t)$ is also a valid solution. Note that this is not true for non-conserving forces, or dissipative forces.

Can we then tell if we are either following a path A in reverse, or a path B in forward motion? If the field in question is a magnetic field it might be possible. Imagine an electron spiralling in a magnetic field, it might be possible, from a macroscopic view, to get a sense of rotation by noting the labelling of the magnetic poles. However, if we also reverse the current that generates $\mathbf{B}$, then the situations would be symmetrical. That is, we can say formally that Maxwells equations, e.g.

$$\nabla \cdot \mathbf{E}, \quad \nabla \times \mathbf{B} - \frac{1}{c}\partial_t \mathbf{E} = \frac{4\pi\mathbf{j}}{c}, \quad \nabla \times \mathbf{E} = -\frac{1}{c}\partial_t \mathbf{B}, \tag{4.96}$$

and the Lorentz force

$$\mathbf{F} = e\left[\mathbf{E} + \frac{1}{c}(\mathbf{v}\times\mathbf{B})\right] \tag{4.97}$$

are invariant under the transformation $t \to -t$, if we also have

$$\mathbf{E}\to\mathbf{E}, \quad \mathbf{B}\to-\mathbf{B}, \quad \rho\to\rho, \quad \mathbf{j}\to-\mathbf{j}, \quad \text{and} \quad \mathbf{v}\to-\mathbf{v}. \tag{4.98}$$

Moving away from the classical point of view, and instead looking to wave mechanics, the Schrödinger equation reads

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi. \tag{4.99}$$

If we suppose that $\psi(\mathbf{x},t)$ is a solution, then it is easy to see that $\psi(\mathbf{x},-t)$ is not a solution due to the first time derivative. We do see however, that $\psi^*(\mathbf{x},-t)$ is a solution. For an energy eigenstate

$$\psi(\mathbf{x},t) = u_n(\mathbf{x})e^{-iE_nt/\hbar}, \quad \text{and} \quad \psi^*(\mathbf{x},-t) = u_n^*(\mathbf{x})e^{-iE_nt/\hbar}. \tag{4.100}$$

We conjecture that time reversal and complex conjugation are related.

### 4.4.1 Digression on Symmetry Operations

Consider a symmetry operation

$$|\alpha\rangle \to |\tilde{\alpha}\rangle \quad \text{and} \quad |\beta\rangle \to \left|\tilde{\beta}\right\rangle. \tag{4.101}$$

It is reasonable to think that the inner product, $\langle\beta|\alpha\rangle = \left\langle\tilde{\beta}\middle|\tilde{\alpha}\right\rangle$, is conserved. For symmetry operations where the corresponding operator is unitary, that is, for rotations, translations, parity, this is true.

We not that when considering time reversal, this requirement is too strong, and we instead make a weaker statement

$$\left|\left\langle\tilde{\beta}\middle|\tilde{\alpha}\right\rangle\right| = |\langle\beta|\alpha\rangle|. \tag{4.102}$$

This statement can also be satisfied if

$$\left\langle\tilde{\beta}\middle|\tilde{\alpha}\right\rangle = \langle\beta|\alpha\rangle^* = \langle\alpha|\beta\rangle. \tag{4.103}$$

**Definition** The transformation

$$|\alpha\rangle \to |\tilde{\alpha}\rangle = \theta|\alpha\rangle \quad \text{and} \quad |\beta\rangle \to \left|\tilde{\beta}\right\rangle = \theta|\beta\rangle \tag{4.104}$$

is said to be antiunitary if

$$\left\langle\tilde{\beta}\middle|\tilde{\alpha}\right\rangle = \langle\beta|\alpha\rangle^*, \tag{4.105}$$

and

$$\theta(c_1|\alpha\rangle + c_2|\beta\rangle) = c_1^*\theta|\alpha\rangle + c_2^*\theta|\beta\rangle. \tag{4.106}$$

The operator $\theta$ is an antiunitary operator.

We claim that an antiunitary operator can be written on the form

$$\theta = UK, \tag{4.107}$$

where $U$ is a unitary operator and $K$ is the complex conjugate operator, defined such that it complex conjugates any coefficient multiplying a ket. We have for $K$

$$Kc|\alpha\rangle = c^* K|\alpha\rangle \tag{4.108}$$

Suppose we expand $|\alpha\rangle$ in terms of $\{|a'\rangle\}$: $|\alpha\rangle = \sum_{a'} |a'\rangle\langle a'|\alpha\rangle$, then

$$K|\alpha\rangle = K\sum_{a'}\langle a'|\alpha\rangle|a'\rangle \tag{4.109a}$$

$$= \sum_{a'}\langle a|\alpha\rangle^* K|a'\rangle \tag{4.109b}$$

$$= \sum_{a'}\langle a|\alpha\rangle^*|a'\rangle. \tag{4.109c}$$

We note that $K$ does not change the base ket, as its explicit representation is

$$|a'\rangle = (0, 0, \cdots, 0, 1, 0, \cdots, 0, 0)^T. \tag{4.110}$$

However, if we take spin as an example. Suppose the $S_y$ eigenkets in the basis of $S_z$, then

$$K\left(\frac{1}{\sqrt{2}}|\uparrow\rangle \pm \frac{i}{\sqrt{2}}|\downarrow\rangle\right) = \frac{1}{\sqrt{2}}|\uparrow\rangle \mp \frac{i}{\sqrt{2}}|\downarrow\rangle. \tag{4.111}$$

If the $S_y$ eigenkets were themselves the basis, then $K$ would not affect them. That is, the effect of $K$ depend on the basis.

### 4.4.2 Time-Reversal Operator

We will denote the time-reversal operator by $\Theta$, and then consider the time reversed state $\Theta|\alpha\rangle$. A more appropriate name for $\Theta|\alpha\rangle$ would be the motion-reversed state. E.g. we expect $\Theta|\mathbf{p}'\rangle = |-\mathbf{p}'\rangle$.

Consider a system in state $|\alpha\rangle$ at time $t_0 = 0$. At a later time $t = \delta t$, the system is in state

$$|\alpha, 0; \delta t\rangle = \left(1 - \frac{iH}{\hbar}\delta t\right)|\alpha\rangle \tag{4.112}$$

where $H$ is the Hamiltonian that characterizes the time evolution. Now suppose that before the time evolution, at $t_0 = 0$ we apply the time-reversal operator, then, at time $t = \delta t$ we have

$$\left(1 - \frac{iH}{\hbar}\delta t\right)\Theta|\alpha\rangle. \tag{4.113}$$

If motion obeys symmetry under time reversal, then this ket is the same as

$$\Theta|\alpha, 0; -\delta t\rangle, \tag{4.114}$$

that is, we consider a state ket at an earlier time $t = -\delta t$, and then reverse $\mathbf{p}$ and $\mathbf{J}$. Formally we write

$$\left(1 - \frac{iH}{\hbar}\delta t\right)\Theta|\alpha\rangle = \Theta\left(1 - \frac{iH}{\hbar}(-\delta t)\right)|\alpha\rangle, \tag{4.115}$$

and if true for any ket, then
$$-iH\Theta|\cdot\rangle = \Theta iH|\cdot\rangle. \tag{4.116}$$

Note that $\Theta$ cannot be unitary as that would make the cancellation of $i$ in the above equation valid, yielding
$$-H\Theta = \Theta H. \tag{4.117}$$
If we now consider an energy eigenket $|n\rangle$ with eigenvalue $E_n$, then the time-reversed state $\Theta|n\rangle$ would have the energy eigenvalue
$$H\Theta|n\rangle = -\Theta H|n\rangle = (-E_n)\Theta|n\rangle. \tag{4.118}$$
Thus, it implies that the time reversed energy state has energy eigenvalues $-E_n$, which is nonsensical even in the most elementary cases. Consider a free particle with Hamiltonian $H = \mathbf{p}^2/2m$. Possible eigenvalues would be positive semidefinite, not negative. If the time-reversal operator was unitary we would need $\mathbf{p}^2 \to -\mathbf{p}^2$ and $\mathbf{p} \to -\mathbf{p}$ which is not physical.

That is, for time reversal to be a useful symmetry we need the operator $\Theta$ to be antiunitary, and thus
$$\Theta iH|\cdot\rangle = -i\Theta H|\cdot\rangle, \tag{4.119}$$
allowing us to now cancel the $i$, giving
$$\Theta H = H\Theta. \tag{4.120}$$
The above equation is the fundamental property of the Hamiltonian under time reversal.

Since $\Theta$ is an antilinear operator, some part of the braket notation might be confusing, as it was made for linear operators. We only ever mean $\Theta$ to act on kets to the right, and never on bras to the left. That is,
$$\langle\beta|\Theta|\alpha\rangle = \langle\beta|\cdot(\Theta|\alpha\rangle) \neq (\langle\beta|\Theta)\cdot|\alpha\rangle. \tag{4.121}$$

We recall the action of $\Theta$ on kets
$$|\tilde{\alpha}\rangle = \Theta|\alpha\rangle \quad \text{and} \quad \left|\tilde{\beta}\right\rangle = \Theta|\beta\rangle. \tag{4.122}$$
However, it is often also convenient to discuss if operators are odd or even under time reversal. For a linear operator $X$, let us define
$$|\gamma\rangle = X^\dagger|\beta\rangle, \tag{4.123}$$
then by dual correspondence
$$\langle\gamma| = \langle\beta|X. \tag{4.124}$$
Now,
$$\begin{aligned}
\langle\beta|X|\alpha\rangle &= \langle\gamma|\alpha\rangle && (4.125a)\\
&= \langle\tilde{\alpha}|\tilde{\gamma}\rangle && (4.125b)\\
&= \left\langle\tilde{\alpha}\middle|\Theta X^\dagger\middle|\beta\right\rangle && (4.125c)\\
&= \left\langle\tilde{\alpha}\middle|\Theta X^\dagger\Theta^{-1}\Theta\middle|\beta\right\rangle && (4.125d)\\
&= \left\langle\tilde{\alpha}\middle|\Theta X^\dagger\Theta^{-1}\middle|\tilde{\beta}\right\rangle, && (4.125e)
\end{aligned}$$

which is an important result. Notably, for Hermitian observables $A$ we get

$$\langle\beta|A|\alpha\rangle = \left\langle\tilde{\alpha}\middle|\Theta A\Theta^{-1}\middle|\tilde{\beta}\right\rangle. \tag{4.126}$$

We call observables even (+) or odd (−) depending on the sign in

$$\Theta A\Theta^{-1} = \pm A. \tag{4.127}$$

It is possible to obtain a phase restriction on the matrix elements of $A$ taken with respect to time reversed states:

$$\langle\beta|A|\alpha\rangle = \pm\left\langle\tilde{\beta}\middle|A\middle|\tilde{\alpha}\right\rangle^{*}. \tag{4.128}$$

If we have $|\beta\rangle = |\alpha\rangle$, then we have an expectation value, nd

$$\langle\alpha|A|\alpha\rangle = \pm\langle\tilde{\alpha}|A|\tilde{\alpha}\rangle. \tag{4.129}$$

We will use $\mathbf{p}$ as an example, looking at its expectation value. The expectation value of $\mathbf{p}$ should reasonably be of opposite sign under time reversal, that is

$$\langle\alpha|\mathbf{p}|\alpha\rangle = -\langle\tilde{\alpha}|\mathbf{p}|\tilde{\alpha}\rangle \tag{4.130}$$

so $\mathbf{p}$ is an odd operator

$$\Theta\mathbf{p}\Theta^{-1} = -\mathbf{p}. \tag{4.131}$$

This gives

$$\begin{aligned}
\mathbf{p}\Theta|\mathbf{p}'\rangle &= -\Theta\mathbf{p}\Theta^{-1}\Theta|\mathbf{p}'\rangle && (4.132\text{a})\\
&= -\Theta\mathbf{p}|\mathbf{p}'\rangle && (4.132\text{b})\\
&= (-\mathbf{p}')\Theta|\mathbf{p}'\rangle, && (4.132\text{c})
\end{aligned}$$

so the eigenvalue of the time reversed momentum ket is $-\mathbf{p}'$. Similarly, from the reasonable requirement

$$\langle\alpha|\mathbf{x}|\alpha\rangle = \langle\tilde{\alpha}|\mathbf{x}|\tilde{\alpha}\rangle, \tag{4.133}$$

we get that

$$\begin{aligned}
\Theta\mathbf{x}\Theta^{-1} &= \mathbf{x} && (4.134\text{a})\\
\Theta|\mathbf{x}'\rangle &= |\mathbf{x}'\rangle. && (4.134\text{b})
\end{aligned}$$

Not that the above is true up to a global phase.

Consider the canonical commutation relation applied to a general ket

$$[x_i, p_j]|\cdot\rangle = i\hbar\delta_{ij}|\cdot\rangle, \tag{4.135}$$

and then apply $\Theta$ from the left

$$\Theta[x_i, p_j]\Theta^{-1}\Theta|\cdot\rangle = \Theta i\hbar\delta_{ij}|\cdot\rangle \tag{4.136}$$

which gives

$$[x_i, -p_j]\Theta|\cdot\rangle = -i\hbar\delta_{ij}\Theta|\cdot\rangle. \tag{4.137}$$

Note that the antiunitarity of the time-reversal operator conserves the commutation relation $[x_i, p_j] = i\hbar\delta_{ij}$, and thus the canonical commutation relation is invariant under time reversal. We could have used this to require $\Theta$ to be antiunitary instead. Similarly, to preserve

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k, \tag{4.138}$$

the angular-momentum operator must be odd under time reversal, that is

$$\Theta\mathbf{J}\Theta^{-1} = -\mathbf{J}. \tag{4.139}$$

Note that for a spinless system $\mathbf{J} = \mathbf{x}\times\mathbf{p}$.

### 4.4.3 Wave Function

Consider a spinless single-particle system at time $t = 0$ in the state $|\alpha\rangle$. In the expansion in the position representation, the wave function $\langle \mathbf{x}'|\alpha\rangle$ appears as the coefficient

$$\alpha = \int d^3x' |\mathbf{x}'\rangle\langle \mathbf{x}'|\alpha\rangle. \tag{4.140}$$

Applying the time reversal operator

$$\Theta|\alpha\rangle = \Theta \int d^3x' \langle \mathbf{x}'|\alpha\rangle|\mathbf{x}'\rangle \tag{4.141a}$$

$$= \int d^3x' \langle \mathbf{x}'|\alpha\rangle^* \Theta|\mathbf{x}'\rangle \tag{4.141b}$$

$$= \int d^3x' \langle \mathbf{x}'|\alpha\rangle^* |\mathbf{x}'\rangle, \tag{4.141c}$$

where we choose the phase such that $\Theta|\mathbf{x}'\rangle = |\mathbf{x}'\rangle$. We have recovered the earlier rule

$$\psi(\mathbf{x}') \xrightarrow{\Theta} \psi^*(\mathbf{x}'). \tag{4.142}$$

The angular part of the wave function is given by the spherical harmonic $Y_l^m$, and with our normal phase convention we find

$$Y_l^m(\vartheta, \varphi) \xrightarrow{\Theta} Y_l^{m*}(\vartheta, \varphi) = (-1)^m Y_l^{-m}(\vartheta, \varphi). \tag{4.143}$$

Thus, we find that

$$\Theta|l, m\rangle = (-1)^m|l, -m\rangle. \tag{4.144}$$

Time reversal, reverses the direction of the probability flux for a wave function since $m$ catches a sign flip.

**Theorem** Suppose the Hamiltonian is invariant under time reversal and the energy eigenket $|n\rangle$ is nondegenerate; then the corresponding energy eigenfunction is real (or, more generally, a real function times a phase factor independent of $\mathbf{x}$).

**Proof** We note that

$$H\Theta|n\rangle = \Theta H|n\rangle = E_n\Theta|n\rangle, \tag{4.145}$$

and thus $|n\rangle$ and the time reversed state $\Theta|n\rangle$ has the same energy. The nondegeneracy assumption helps us make the conclusion that $|n\rangle$ and $\Theta|n\rangle$ is the same state. The wave functions must therefore be identical:

$$\langle \mathbf{x}'|n\rangle = \langle \mathbf{x}'|n\rangle^*. \tag{4.146}$$

That is, the wave function must be real, up to a phases factor independent of $\mathbf{x}$. □

Thus, a nondegenerate bound state always have a real wave function. For the case of the hydrogen atom, we have a degeneracy, and thus the wave function is complex, namely $Y_l^m$ is complex. A plane wave is also degenerate, and is therefore also complex.

For a spinless system, the position wave function for a time reversed state is simply obtained by complex conjugation. If we instead consider the momentum wave function, we see a different result. Consider

$$\Theta|\alpha\rangle = \Theta \int d^3p' \langle \mathbf{p}'|\alpha\rangle |\mathbf{p}'\rangle \tag{4.147a}$$

$$= \int d^3p' \langle \mathbf{p}'|\alpha\rangle^* \Theta|\mathbf{p}'\rangle \tag{4.147b}$$

$$= \int d^3p' \langle \mathbf{p}'|\alpha\rangle^* |-\mathbf{p}'\rangle \tag{4.147c}$$

$$= \int d^3p' \langle -\mathbf{p}'|\alpha\rangle^* |\mathbf{p}'\rangle, \tag{4.147d}$$

where we do a substitution in the last step. Thus, for time reversal of the momentum wave function, we identify $\phi^*(-\mathbf{p}')$ as the momentum wave function for the time reversed state.

### 4.4.4 Time Reversal for a Spin $\frac{1}{2}$ System

Consider now a particle with spin $\frac{1}{2}$. We can write the eigenket of $\mathbf{S}\cdot\hat{\mathbf{n}}$ with eigenvalue $\hbar/2$ as

$$|\hat{\mathbf{n}};\uparrow\rangle = e^{-iS_z\alpha/\hbar} e^{-iS_y\beta/\hbar}|\uparrow\rangle, \tag{4.148}$$

where $\hat{\mathbf{n}}$ is characterized by the polar angle $\beta$ and azimuthal angle $\alpha$. Since angular momentum operators are odd under time reversal we have

$$\Theta|\hat{\mathbf{n}};\uparrow\rangle = e^{-iS_z\alpha/\hbar} e^{-iS_y\beta/\hbar}\Theta|\uparrow\rangle = \eta|\hat{\mathbf{n}};\downarrow\rangle. \tag{4.149}$$

It is also easy to see that

$$|\hat{\mathbf{n}};\downarrow\rangle = e^{-iS_z\alpha/\hbar} e^{-iS_y(\beta+\pi)/\hbar}|\uparrow\rangle. \tag{4.150}$$

Decomposing the time reversal operator $\Theta = UK$, and comparing the previous equations we see that

$$\Theta = \eta e^{-iS_y\pi/\hbar}K = i\eta\left(\frac{2S_y}{\hbar}\right)K, \tag{4.151}$$

where $\eta$ is an arbitrary phase.

Note that

$$e^{-iS_y\pi/\hbar}|\uparrow\rangle = -i\frac{2}{\hbar}S_y|\uparrow\rangle \tag{4.152a}$$

$$= -i\frac{2}{\hbar}\left(\frac{i\hbar}{2}\right)|\downarrow\rangle \tag{4.152b}$$

$$= |\downarrow\rangle \tag{4.152c}$$

and

$$e^{-iS_y\pi/\hbar}|\downarrow\rangle = -i\frac{2}{\hbar}S_y|\downarrow\rangle \tag{4.153a}$$

$$= -i\frac{2}{\hbar}\left(\frac{-i\hbar}{2}\right)|\uparrow\rangle \tag{4.153b}$$

$$= -|\uparrow\rangle. \tag{4.153c}$$

For a general spin $\frac{1}{2}$ ket, we have

$$\Theta(c_\uparrow|\uparrow\rangle + c_\downarrow|\downarrow\rangle) = \eta c_\uparrow^*|\downarrow\rangle - \eta c_\downarrow^*|\uparrow\rangle. \tag{4.154}$$

Applying $\Theta$ again,

$$\Theta^2(c_\uparrow|\uparrow\rangle + c_\downarrow|\downarrow\rangle) = \Theta(\eta c_\uparrow^*|\downarrow\rangle - \eta c_\downarrow^*|\uparrow\rangle) \tag{4.155a}$$

$$= \eta^* c_\uparrow \Theta|\downarrow\rangle - \eta^* c_\downarrow|\uparrow\rangle \tag{4.155b}$$

$$= -|\eta|^2 c_\uparrow|\uparrow\rangle - |\eta|^2 c_\downarrow|\downarrow\rangle \tag{4.155c}$$

$$= -(c_\uparrow|\uparrow\rangle + c_\downarrow|\downarrow\rangle), \tag{4.155d}$$

that is

$$\Theta^2 = -1, \tag{4.156}$$

for any spin orientation. This is true regardless of what phase we choose. This is very interesting, especially when comparing for a spinless state, where

$$\Theta^2 = 1. \tag{4.157}$$

Generally, we can prove that

$$\Theta^2|j \text{ half-integer}\rangle = -|j \text{ half-integer}\rangle \tag{4.158}$$

and

$$\Theta^2|j \text{ integer}\rangle = |j \text{ integer}\rangle, \tag{4.159}$$

and thus the eigenvalue of $\Theta^2$ is given by $(-1)^{2j}$. We can generalize the time reversal operator for any $j$, as

$$\Theta = \eta e^{-iJ_y\pi/\hbar}K. \tag{4.160}$$

For a general ket $\alpha$ expanded in the basis of $|j,m\rangle$ we have

$$\Theta(\Theta\sum_m|j,m\rangle\langle j,m|\alpha\rangle) = \Theta(\eta\sum_m e^{-iJ_y\pi/\hbar}|j,m\rangle\langle j,m|\alpha\rangle^*) \tag{4.161a}$$

$$= |eta|^2\sum_m e^{-2iJ_y\pi/\hbar}|j,m\rangle\langle j,m|\alpha\rangle \tag{4.161b}$$

$$= (-1)^{2j}\sum_m|j,m\rangle\langle j,m|\alpha\rangle \tag{4.161c}$$

$$= (-1)^{2j}|\alpha\rangle. \tag{4.161d}$$

For $|j$ integer$\rangle$, can be any even number electron state, while $|j$ half-integer$\rangle$ can be any odd number electron state. Note that they need not be eigenstates of $\mathbf{J}^2$.

For the phase convention, a common convention is to take $\eta = i$, such that

$$\Theta|j,m\rangle = i^{2m}|j,-m\rangle, \tag{4.162}$$

regardless of whether $j$ refers to spin, or orbital angular momentum, and if it is integer or half-integer. This is not the only convention, however.

For a spherical tensor $T_q^{(k)}$, we call it even or odd depending on

$$\Theta T_{q=0}^{(k)}\Theta^{-1} = \pm T_{q=0}^{(k)} \tag{4.163}$$

and for the expectation value we find

$$\left\langle \alpha,j,m\middle|T_0^{(k)}\middle|\alpha,j,m\right\rangle = \pm(-1)^k\left\langle \alpha,j,m\middle|T_0^{(k)}\middle|\alpha,j,m\right\rangle. \tag{4.164}$$

# 6 Scattering Theory

Scattering processes are the processes that transform an initial state into a final state, by some potential, which we will treat as a time-dependent perturbation. This is how we learn about experimental processes.

## 6.1 Scattering as a Time-Dependent Perturbation

We make the assumption that the Hamiltonian may be written as

$$H = H_0 + V(\mathbf{x}), \tag{6.1}$$

with

$$H_0 = \frac{\mathbf{p}^2}{2m}, \tag{6.2}$$

the kinetic energy operator. The eigenvalues of $H_0$ are

$$E_{\mathbf{k}} = \frac{\hbar^2\mathbf{k}^2}{2m}. \tag{6.3}$$

Plane wave eigenvectors of $H_0$, with eigenvalues $E_{\mathbf{k}}$, are $|\mathbf{k}\rangle$, and we make the assumption that the scattering potential $V(\mathbf{r})$ is time independent. We will treat this as an incoming particle seeing the scattering potential which is active only when the particle is close to the scatterer. This analysis will be performed using time-dependent perturbation theory in the interaction picture.

We know that the state evolution is

$$|\alpha, t_0; t\rangle_I = U_I(t, t_0)|\alpha, t_0; t_0\rangle, \tag{6.4}$$

where $U_I(t, t_0)$ is the time evolution operator in the interaction picture, satisfying

$$i\hbar\partial_t U_I(t, t_0) = V_I(t)U_I(t, t_0). \tag{6.5}$$

We have $U_I(t_0, t_0) = 1$ and $V_I(t) = \exp(iH_0t/\hbar)V\exp(-iH_0t/\hbar)$, and the exact solution to the equation may be written as

$$U_I(t, t_0) = 1 - \frac{i}{\hbar}\int_{t_0}^{t} dt' V_I(t')U_I(t', t_0). \tag{6.6}$$

The transition amplitude for the transformation from an initial state $|i\rangle$ into a final state $|n\rangle$, both eigenstate to $H_0$, can be written as

$$\langle n|U_I(t, t_0)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\sum_m \langle n|V|m\rangle \int_{t_0}^{t} dt' e^{i\omega_{nm}t'}\langle m|U_I(t', t_0)|i\rangle, \tag{6.7}$$

where $\hbar\omega_{nm} = E_n - E_m$ and $\delta ni = \langle n|i\rangle$. We arrive here by inserting the completeness relation $\sum_m |m\rangle\langle m|$, and converting $V_I \to V$.

In scattering theory we deal with a continuum of states, not discrete ones, we therefore have to adjust the above equation to account for this. We do this by quantizing the scattering states in a cubic volume of side $L$. We have, in the coordinate representation, the wave function

$$\langle \mathbf{x}|\mathbf{k}\rangle = \frac{1}{L^{3/2}}e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{6.8}$$

where we will take $L \to \infty$ at the end. Note that $\langle \mathbf{k'}|\mathbf{k}\rangle = \delta_{\mathbf{k'k}}$.

The initial and final states exist only asymptomatically, and we therefore have to have $t \to \infty$ and $t_0 \to -\infty$.

We start by doing a first order treatment, where we set $\langle m|U_I(t', t_0)|i\rangle = \delta_{mi}$, which gives

$$\langle n|U_I(t,t_0)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\langle n|V|i\rangle \int_{t_0}^{t} dt' e^{i\omega_{ni}t'}. \tag{6.9}$$

For the $t \to \infty$ we have Fermi's golden rule, and in order to include the behaviour for $t_0 \to -\infty$ we define a matrix $T_{ni}$ by

$$\langle n|U_I t, t_0|i\rangle = \delta_{ni} - \frac{i}{\hbar}T_{ni} \int_{t_0}^{t} dt' e^{i\omega_{ni}t'+\varepsilon t'} \tag{6.10}$$

with $\varepsilon > 0$ and $t \ll 1/\varepsilon$, and we need to make sure to take the limit $\varepsilon \to 0$ before we take $t \to \infty$. We now define the scattering matrix

$$S_{ni} = \lim_{t\to\infty}\left[\lim_{\varepsilon\to 0}\langle n|U_I t, -\infty|i\rangle\right] = \delta_{ni} - \frac{i}{\hbar}T_{ni}\int_{-\infty}^{\infty} dt' e^{i\omega nit'} = \delta_{ni} - 2\pi i\delta(E_n - E_i)T_{ni}. \tag{6.11}$$

We can see that the scattering matrix consists of two parts. One part where the final and initial states are the same, and one part governed by the $T$-matrix, with some sort of scattering.

### 6.1.1 Transition Rates and Cross Sections

We define that transition rate as

$$w(i \to n) = \partial_t |\langle n|U_I t, -\infty|i\rangle|^2 \tag{6.12}$$

and for $|i\rangle \neq |n\rangle$, we have

$$\langle n|U_I(t,-\infty)|i\rangle = -\frac{i}{\hbar}T_{ni}\int_{-\infty}^{t} dt' e^{i\omega_{ni}t'+\varepsilon t'} = -\frac{i}{\hbar}T_{ni}\frac{e^{i\omega_{ni}t+\varepsilon t}}{i\omega_{ni}+\varepsilon}. \tag{6.13}$$

The transition rate for this is then

$$w(i \to n) = \partial_t\left[\frac{1}{\hbar^2}|T_{ni}|^2\frac{e^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2}\right] = \frac{1}{\hbar^2}|T_{ni}|^2\frac{2\varepsilon e^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2}. \tag{6.14}$$

Here, we need to take $\varepsilon \to 0$ first, before taking $t \to \infty$. If $\omega_{ni} \neq 0$, we see that $w \to 0$, while for $\omega_{ni} = 0$, we get a delta function. We find that

$$\frac{ee^{2\varepsilon t}}{\omega_{ni}^2+\varepsilon^2} = \pi\delta(\omega_{ni}) = \pi\hbar\delta(E_n - E_i), \tag{6.15}$$

and we get the transmission rate

$$w(i \to n) = \frac{2\pi}{\hbar}|T_{ni}|^2\delta(E_n - E_i) \tag{6.16}$$

which is independent of time so the limit $t \to \infty$ is trivial. This is very similar to Fermi's golden rule, but with $T_{ni}$ instead of $V_{ni}$, which is a more general matrix. We can use this transition rate to determine a scattering cross section.

Similarly to Fermi's golden rule, we need to determine the density of final states, $\rho(E_n) = \Delta n/\Delta E_n$, so that we can integrate over the final state energy. We will consider elastic scattering, where $|i\rangle = |\mathbf{k}\rangle, |n\rangle = |\mathbf{k}'\rangle$, and $|\mathbf{k}| = |\mathbf{k}'| = k$. Using the normalization due to our cubic volume, we can write

$$E_{\mathbf{n}} = \frac{\hbar^2 \mathbf{k}'^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{2\pi}{L}\right)^2 |\mathbf{n}|^2, \tag{6.17}$$

and thus

$$\Delta E_{\mathbf{n}} = \partial_{\mathbf{n}}(\Delta E_{\mathbf{n}})\Delta|\mathbf{n}| = \frac{\hbar^2}{m}\left(\frac{2\pi}{L}\right)^2 |\mathbf{n}|\Delta|\mathbf{n}|, \tag{6.18}$$

where $\mathbf{n} = n_x\hat{\mathbf{i}} + n_y\hat{\mathbf{j}} + n_z\hat{\mathbf{k}}$ and $n_{x,y,z} \in \mathbb{Z}$. We have $\mathbf{n} = (L/2\pi)|\mathbf{k}'| = (L/2\pi)k$ and $L$ is large, so we can imagine $|\mathbf{n}|$ as almost continuous. The number of states within a spherical shell of radius $|\mathbf{n}|$ and thickness $\Delta|\mathbf{n}|$ is

$$\Delta n = 4\pi|\mathbf{n}|^2\Delta|\mathbf{n}| \times \frac{d\Omega}{4\pi}, \tag{6.19}$$

considering the fraction of solid angle represented by the final state vector $\mathbf{k}'$. Thus,

$$\rho(E_n) = \frac{\Delta n}{\Delta E_n} = \frac{m}{\hbar^2}\left(\frac{L}{2\pi}\right)^2 |\mathbf{n}|d\Omega = \frac{mk}{\hbar^2}\left(\frac{L}{2\pi}\right)^3 d\Omega. \tag{6.20}$$

Integrating over the final states gives

$$w(i \to n) = \frac{mkL^3}{(2\pi)^2\hbar^3}|T_{ni}|^2 d\Omega. \tag{6.21}$$

This concept of cross section is used to understand the transition rate in scattering experiments. Determining the rate at which particles scatter into a solid angle $d\Omega$ from a beam of particles with momentum $\hbar\mathbf{k}$. The speed of the particles is $v = \hbar k/m$, and the time needed to cross the cubic volume is $t = L/v$. The particle flux in the beam is $v/L^3$, and the probability flux for the wave function becomes

$$\mathbf{j}(\mathbf{x}, t) = \left(\frac{\hbar}{m}\right)\frac{\mathbf{k}}{L^3} = \frac{v}{L^3}. \tag{6.22}$$

The cross section $d\sigma$ is defined as the transition rate divided by the flux. Finally, we arrive at

$$\frac{d\sigma}{d\Omega} = \left(\frac{mL^3}{2\pi\hbar^2}\right)^2 |T_{ni}|^2. \tag{6.23}$$

We now have to relate the matrix elements $T_{ni}$ to the scattering potential $V(\mathbf{r})$.

### 6.1.2 Solving for the $T$ matrix

Recalling the definition of the $T$ matrix we have

$$\langle n|U_I(t, -\infty)|i\rangle = \delta_{ni} + \frac{1}{\hbar}T_{ni}\frac{e^{i\omega_{ni}t+\varepsilon t}}{-\omega_{ni} + i\varepsilon} \tag{6.24}$$

and

$$\langle n|U_I(t, -\infty)|i\rangle = \delta_{ni} - \frac{i}{\hbar}\sum_m V_{nm}\int_{-\infty}^{t} dt' e^{i\omega_{nm}t'}\langle m|U_I(t', -\infty)|i\rangle. \tag{6.25}$$

Inserting the former equation into the latter under the integral sign we obtain three terms. The first of which is $\delta_{ni}$, the second looks like the former equation but with $T_{ni}$ replaced by $V_{ni}$. The final term is

$$-\frac{i}{\hbar}\frac{1}{\hbar}\sum_m V_{nm}\frac{T_{mi}}{-\omega_{mi}+i\varepsilon}\int_{-\infty}^{t} dt' e^{i(\omega_{nm}+\omega_{mi})t'+\varepsilon t'}. \tag{6.26}$$

Since $\omega_{nm}+\omega_{mi}=\omega_{ni}$, the result of the integral can be taken out from the summation. Gathering terms and comparing to the former equation we find that

$$T_{ni} = V_{ni} + \frac{1}{\hbar}\sum_m V_{nm}\frac{T_{mi}}{-\omega_{mi}+i\varepsilon} = V_{ni} + \sum_m V_{nm}\frac{T_{mi}}{E_i - E_m + i\hbar\varepsilon}. \tag{6.27}$$

This is an inhomogeneous system of linear equation which can be solved for $T_{ni}$ in terms of the matrix elements $V_{nm}$, which are known.

We can define a set of vectors $\left|\psi^{(+)}\right\rangle$ in terms of components in some basis $|j\rangle$, such that

$$T_{ni} = \sum_j \langle n|V|j\rangle\left\langle j\middle|\psi^{(+)}\right\rangle = \left\langle n\middle|V\middle|\psi^{(+)}\right\rangle. \tag{6.28}$$

Inserting this into the equation for $T_{ni}$, we find

$$\left\langle n\middle|V\middle|\psi^{(+)}\right\rangle = \langle n|V|i\rangle + \sum_m \langle n|V|m\rangle\frac{\left\langle m\middle|V\middle|\psi^{(+)}\right\rangle}{E_i - E_m + i\hbar\varepsilon}. \tag{6.29}$$

Since this must be true for all $|n\rangle$ we find

$$\left|\psi^{(+)}\right\rangle = |i\rangle + \frac{1}{E_i - H_0 + i\hbar\varepsilon}V\left|\psi^{(+)}\right\rangle, \tag{6.30}$$

which is known as the Lippmann-Schwinger equation. We can define an operator $T$ with matrix elements $T_{ni}$ by writing

$$T|i\rangle = V\left|\psi^{(+)}\right\rangle, \tag{6.31}$$

we can then find the operator equation

$$T = V + V\frac{1}{E_i - H_0 + i\hbar\varepsilon}T. \tag{6.32}$$

### 6.1.3 Scattering from the Future to the Past

We could instead imagine the scattering process evolving backwards in time, from a plane wave state $|i\rangle$ in the far future, to a state $|n\rangle$ in the distant past. We would in this case write

$$U_I(t,t_0) = 1 + \frac{i}{\hbar}\int_t^{t_0} dt' V_I(t')U_I(t',t_0) \tag{6.33}$$

and the $T$ matrix is the n defined as

$$\langle n|U_I(t,t_0)|i\rangle = \delta_{ni} + \frac{1}{\hbar}T_{ni}\int_t^{t_0} dt' e^{i\omega_{ni}t'-\varepsilon t'}. \tag{6.34}$$

In this case we can define $T$ using a different set of states $\left|\psi^{(-)}\right\rangle$ through

$$T|i\rangle = V\left|\psi^{(-)}\right\rangle. \tag{6.35}$$

## 6.2 The Scattering Amplitude

We make the substitution $\hbar\varepsilon \to \varepsilon$ in the Lippmann-Schwinger equation. We continue considering elastic scattering, and we use $E$ for the initial (and final) energy. Likewise, we write

$$\big|\psi^{(\pm)}\big\rangle = |i\rangle + \frac{1}{E - H_0 \pm i\varepsilon} V \big|\psi^{(\pm)}\big\rangle \tag{6.36}$$

and multiply by $\langle\mathbf{x}|$ from the left, and by inserting the completeness relation for position states we get

$$\big\langle\mathbf{x}\big|\psi^{(\pm)}\big\rangle = \langle\mathbf{x}|i\rangle + \int d^3x' \left\langle\mathbf{x}\left|\frac{1}{E - H_0 \pm i\varepsilon}\right|\mathbf{x}'\right\rangle \big\langle\mathbf{x}'\big|V\big|\psi^{(\pm)}\big\rangle, \tag{6.37}$$

which is an integral equation for scattering. The first step is to evaluate the function

$$G_\pm(\mathbf{x}, \mathbf{x}') = \frac{\hbar^2}{2m}\left\langle\mathbf{x}\left|\frac{1}{E - H_0 \pm i\varepsilon}\right|\mathbf{x}'\right\rangle. \tag{6.38}$$

The eigenstates of $H_0$ are easier to evaluate in the momentum basis, and we therefore insert complete sets of these. Doing this, we obtain

$$G_\pm(\mathbf{x}, \mathbf{x}') = \frac{\hbar^2}{2m}\sum_{\mathbf{k}'}\sum_{\mathbf{k}''}\langle\mathbf{x}|\mathbf{k}'\rangle\left\langle\mathbf{k}'\left|\frac{1}{E - H_0 \pm i\varepsilon}\right|\mathbf{k}''\right\rangle\langle\mathbf{k}''|\mathbf{x}'\rangle \tag{6.39}$$

and letting $H_0$ act on $|\mathbf{k}'\rangle$ we find

$$\left\langle\mathbf{k}'\left|\frac{1}{E - H_0 \pm i\varepsilon}\right|\mathbf{k}''\right\rangle = \frac{\delta_{\mathbf{k}'\mathbf{k}''}}{E - (\hbar^2\mathbf{k}''/2m) \pm i\varepsilon}. \tag{6.40}$$

Recalling

$$\langle\mathbf{x}|\mathbf{k}'\rangle = \frac{e^{i\mathbf{k}'\cdot\mathbf{x}}}{L^{3/2}} \tag{6.41}$$

$$\langle\mathbf{k}''|\mathbf{x}'\rangle = \frac{e^{-i\mathbf{k}''\cdot\mathbf{x}'}}{L^{3/2}} \tag{6.42}$$

and writing $E = \hbar^2k^2/2m$, we find

$$G_\pm(\mathbf{x}, \mathbf{x}') = \frac{1}{L^3}\sum_{\mathbf{k}'}\frac{e^{i\mathbf{k}'\cdot(\mathbf{x}-\mathbf{x}')}}{k^2 - k'^2 \pm i\varepsilon}. \tag{6.43}$$

We can convert this to an integral by using

$$\sum_{\mathbf{k}} \to \frac{L^3}{(2\pi)^3}\int d^3k, \tag{6.44}$$

as $L \to \infty$. We now get

$$G_\pm(\mathbf{x}, \mathbf{x}') = \frac{1}{(2\pi)^3}\int d^3k' \frac{e^{i\mathbf{k}'\cdot(\mathbf{x}-\mathbf{x}')}}{k^2 - k'^2 \pm i\varepsilon}. \tag{6.45}$$

This is analytically solvable by switching to spherical coordinates and using contour integration and residuals. Using a contour in either the upper or lower half plane depending on where the singularity is, and applying Cauchy's integral formula, we find

$$G_{\pm}(\mathbf{x},\mathbf{x}') = -\frac{1}{4\pi}\frac{e^{\pm ik|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|} \tag{6.46}$$

We recognize that $G_{\pm}$ is the Green function for the Helmholtz equation

$$(\nabla^2 + k^2)G_{\pm}(\mathbf{x},\mathbf{x}') = \delta^{(3)}(\mathbf{x}-\mathbf{x}'), \tag{6.47}$$

and for $\mathbf{x} \neq \mathbf{x}'$ the eigenvalue equation is solved by $G_{\pm}$

$$H_0 G_{\pm} = E G_{\pm}. \tag{6.48}$$

Inserting this into the equation for the wave function we find

$$\langle \mathbf{x} | \psi^{(\pm)} \rangle = \langle \mathbf{x} | i \rangle - \frac{2m}{\hbar^2}\int d^3x' \frac{e^{\pm ik|\mathbf{x}-\mathbf{x}'|}}{4\pi|\mathbf{x}-\mathbf{x}'|}\langle \mathbf{x}' | V | \psi^{(\pm)} \rangle. \tag{6.49}$$

Note that this consist of two terms. The first term is the incoming wave, while the second is the effect of scattering. Thus, without any scattering, the scattered wave is the same as the incident wave. We can show that the positive solution $\psi^{(+)}$ corresponds to a plane wave plus and outgoing spherical wave, while the negative solution $\psi^{(-)}$ corresponds to a plane wave plus and incoming spherical wave. Which we see as scattering forwards or backwards in time. Usually we are interested in the positive solution in physics.

Consider a local potential $V$, that is, a potential only a function of the position operator $\mathbf{x}$, and diagonal in the position representation. We can write such a local potential as

$$\langle \mathbf{x}' | V | \mathbf{x}'' \rangle = V(\mathbf{x}')\delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{6.50}$$

By inserting a complete set into $\langle \mathbf{x}' | V | \psi^{(\pm)} \rangle$ we get

$$\langle \mathbf{x}' | V | \psi^{(\pm)} \rangle = \int d^3x'' \langle \mathbf{x}' | V | \mathbf{x}'' \rangle \langle \mathbf{x}'' | \psi^{(\pm)} \rangle = V(\mathbf{x}')\langle \mathbf{x}' | \psi^{(\pm)} \rangle, \tag{6.51}$$

which we can use to simplify the equation for the wave function

$$\langle \mathbf{x} | \psi^{(\pm)} \rangle = \langle \mathbf{x} | \psi^{(\pm)} \rangle = \langle \mathbf{x} | i \rangle - \frac{2m}{\hbar^2}\int d^3x' \frac{e^{\pm ik|\mathbf{x}-\mathbf{x}'|}}{4\pi|\mathbf{x}-\mathbf{x}'|}V(\mathbf{x}')\langle \mathbf{x}' | \psi^{(\pm)} \rangle. \tag{6.52}$$

We understand $\mathbf{x}$ as the vector pointing to the observation point, where we measure, or evaluate, the wave function. The vector $\mathbf{x}'$ is the vector pointing to the scatterer from the incoming wave. When dealing with a finite range potential, we are interested in studying the system at a point far outside the nonvanishing area of the potential, i.e. we don't want to put a detector inside the scatter potential. Generally this means that we can set

$$|\mathbf{x}| \gg |\mathbf{x}'|. \tag{6.53}$$

We introduce $r = |\mathbf{x}|$, $r' = |\mathbf{x}'|$, and $\alpha = \angle(\mathbf{x},\mathbf{x}')$. For $r \gg r'$, we write

$$|\mathbf{x}-\mathbf{x}'| = \sqrt{r^2 - 2rr'\cos\alpha + r'^2} \tag{6.54a}$$

$$= r\sqrt{1 - \frac{2r'}{r}\cos\alpha + \frac{r'^2}{r^2}} \tag{6.54b}$$

$$\approx r - \hat{\mathbf{r}}\cdot\mathbf{x}', \tag{6.54c}$$

where

$$\hat{\mathbf{r}} = \frac{\mathbf{x}}{|\mathbf{x}|}, \quad \text{and} \quad \mathbf{k}' = k\hat{\mathbf{r}}. \tag{6.55}$$

With this we write

$$e^{\pm ik|\mathbf{x}-\mathbf{x}'|} \approx e^{\pm ikr} e^{\mp i\mathbf{k}'\cdot\mathbf{x}'}, \tag{6.56}$$

and for large $r$ it is also valid to replace $1/|\mathbf{x} - \mathbf{x}'|$ by $1/r$.

Choosing the initial state as an eigenstate to the free particle Hamiltonian $H_0$, such that $|i\rangle = |\mathbf{k}\rangle$, we can write

$$\begin{aligned} \langle \mathbf{x} | \psi^{(+)} \rangle &\stackrel{r \text{ large}}{\to} \langle \mathbf{x} | \mathbf{k} \rangle - \frac{1}{4\pi} \frac{2m}{\hbar^2} \frac{e^{ikr}}{r} \int d^3x' e^{-i\mathbf{k}'\cdot\mathbf{x}'} V(\mathbf{x}') \langle \mathbf{x}' | \psi^{(+)} \rangle &\text{(6.57a)} \\ &= \frac{1}{L^{3/2}} \left[ e^{i\mathbf{k}\cdot\mathbf{x}} + \frac{e^{ikr}}{r} f(\mathbf{k}', \mathbf{k}) \right]. &\text{(6.57b)} \end{aligned}$$

Here we see clearly that the scattered wave includes both the incident plane wave, and the outgoing spherical wave with amplitude $f(\mathbf{k}', \mathbf{k})$ given by

$$\begin{aligned} f(\mathbf{k}', \mathbf{k}) &= \frac{1}{4\pi} \frac{2m}{\hbar^2} L^3 \int d^3x' \frac{e^{-i\mathbf{k}'\cdot\mathbf{x}'}}{L^{3/2}} V(\mathbf{x}') \langle \mathbf{x}' | \psi^{(+)} \rangle &\text{(6.58a)} \\ &= -\frac{mL^3}{2\pi\hbar^2} \langle \mathbf{k}' | V | \psi^{(+)} \rangle. &\text{(6.58b)} \end{aligned}$$

Similarly, we can show for $|\psi^-\rangle$ that

$$\langle \mathbf{x} | \psi^{(-)} \rangle = \frac{1}{L^{3/2}} \left[ e^{i\mathbf{k}\cdot\mathbf{x}} + \frac{e^{-ikr}}{r} f(\mathbf{k}', \mathbf{k}) \right] \tag{6.59}$$

and

$$f(\mathbf{k}', \mathbf{k}) = -\frac{mL^3}{2\pi\hbar^2} \langle -\mathbf{k}' | V | \psi^{(-)} \rangle. \tag{6.60}$$

We call $f(\mathbf{k}', \mathbf{k})$ the scattering amplitude, and we remark that the differential cross section can be written

$$\frac{d\sigma}{d\Omega} = |f(\mathbf{k}', \mathbf{k})|^2. \tag{6.61}$$

### 6.2.1 Wave Packet Description

This description is not valid for a particle being scattered by a scattering centre if we model the particle as a wave packet approaching a scattering centre. This is hard, but we notice that if the dimensions of the wave packet is much larger than the size of the scatterer, i.e. the range of $V$, the plane wave approach is valid.

### 6.2.2 The Optical Theorem

A useful and fundamental relationship in quantum mechanics is the optics theorem, which relates the imaginary part of the forward scattering amplitude $f(\mathbf{k}, \mathbf{k})$ ($\vartheta = 0$), to the total cross section $\sigma_{\text{tot}} = \int d\Omega (d\sigma/d\Omega)$, and it reads

$$\Im f(\mathbf{k}, \mathbf{k}) = \frac{k\sigma_{\text{tot}}}{4\pi}. \tag{6.62}$$

The proof of this theorem start with the Lippmann-Schwinger equation, with $|i\rangle \to |\mathbf{k}\rangle$

$$\big|\psi^{(+)}\big\rangle = |\mathbf{k}\rangle + \frac{1}{E - H_0 + i\varepsilon} V \big|\psi^{(+)}\big\rangle \tag{6.63}$$

and we can rewrite this to obtain

$$|\mathbf{k}\rangle = \big|\psi^{(+)}\big\rangle - \frac{1}{E - H_0 + i\varepsilon} V \big|\psi^{(+)}\big\rangle. \tag{6.64}$$

We can use this and write

$$\big\langle\mathbf{k}\big|V\big|\psi^{(+)}\big\rangle = \left[\big\langle\psi^{(+)}\big| - \big\langle\psi^{(+)}\big|V\frac{1}{E - H_0 - i\varepsilon}\right]V\big|\psi^{(+)}\big\rangle \tag{6.65a}$$

$$= \big\langle\psi^{(+)}\big|V\big|\psi^{(+)}\big\rangle - \left\langle\psi^{(+)}\middle|V\frac{1}{E - H_0 - i\varepsilon}V\middle|\psi^{(+)}\right\rangle. \tag{6.65b}$$

We want to take the imaginary part of both sides of this expression. The first term on the right is the expectation value of a Hermitian operator and is thus real. To calculate the imaginary part of the second term we need to use complex analysis and Cauchy's principle value as we get a singularity when $\varepsilon \to 0$. Since $\varepsilon > 0$, we use a contour integral with a small semicircle extending down to the lower half plane around the singularity. Let this semicircle have the radius $\delta$, and let the singularity be at $z_0 = x_0 + i\varepsilon$. To show the trick we will use, let us consider a complex function $f(z) = x + iy$, and we can then write

$$\int_{-\infty}^{\infty} dx \frac{f(x)}{x - x_0} = \mathscr{P}\int_{-\infty}^{\infty} dx \frac{f(x)}{x - x_0} + \int_c dz \frac{f(z)}{z - z_0}, \tag{6.66}$$

where $c$ denotes the semicircle contour, and the Cauchy principle value is defined as

$$\mathscr{P}\int_{-\infty}^{\infty} dx \frac{f(x)}{x - x_0} = \lim_{\delta\to 0}\left\{\int_{-\infty}^{x_0-\delta} dx \frac{f(x)}{x - x_0} + \int_{x_0+\delta}^{\infty} dx \frac{f(x)}{x - x_0}\right\}. \tag{6.67}$$

For the second term we can use calculus of residues to evaluate

$$\int_c dz \frac{f(z)}{z - z_0} = i\pi f(x_0) \quad \text{as } \delta \to 0. \tag{6.68}$$

We now have

$$\int_{-\infty}^{\infty} dx \frac{f(x)}{x - x_0} = \mathscr{P}\int_{-\infty}^{\infty} dx \frac{f(x)}{x - x_0} + i\pi f(x_0). \tag{6.69}$$

Returning to the equation at hand, we start by writing

$$\lim_{\varepsilon\to 0}\left[\frac{1}{E - H_0 - i\varepsilon}\right] = \lim_{\varepsilon\to 0}\int_{-\infty}^{\infty} dE' \frac{\delta(E - E')}{E' - H_0 - i\varepsilon} \tag{6.70a}$$

$$= i\pi\delta(E - H_0), \tag{6.70b}$$

using what we showed earlier, and we have omitted the principle value since it is a real number. Recalling $T|\mathbf{k}\rangle = V\big|\psi^{(+)}\big\rangle$, we can write

$$\Im\big\langle\mathbf{k}\big|V\big|\psi^{(+)}\big\rangle = -\pi\big\langle\mathbf{k}\big|T^\dagger\delta(E - H_0)T\big|\mathbf{k}\big\rangle. \tag{6.71}$$

Taking the imaginary part of the forward scattering amplitude gives

$$\Im f(\mathbf{k},\mathbf{k}) = -\frac{mL^3}{2\pi\hbar^2}\Im\big\langle\mathbf{k}\big|V\big|\psi^{(+)}\big\rangle \tag{6.72a}$$
$$= \frac{mL^3}{2\hbar^2}\big\langle\mathbf{k}\big|T^\dagger\delta(E-H_0)T\big|\mathbf{k}\big\rangle \tag{6.72b}$$
$$= \frac{mL^3}{2\hbar^2}\sum_{\mathbf{k}'}\big\langle\mathbf{k}\big|T^\dagger\delta(E-H_0)\big|\mathbf{k}'\big\rangle\langle\mathbf{k}'|T|\mathbf{k}\rangle \tag{6.72c}$$
$$= \frac{mL^3}{2\hbar^2}\sum_{\mathbf{k}'}\left|\langle\mathbf{k}'|T|\mathbf{k}\rangle\right|^2\delta_{E,\frac{\hbar^2\mathbf{k}'^2}{2m}}, \tag{6.72d}$$

where $E=\hbar^2k^2/2m$. To simplify, we start with

$$\langle\mathbf{k}'|T|\mathbf{k}\rangle = \big\langle\mathbf{k}'\big|V\big|\psi^{(+)}\big\rangle = -\frac{2\pi\hbar^2}{mL^3}f(\mathbf{k}',\mathbf{k}), \tag{6.73}$$

and using

$$\sum_{\mathbf{k}} \to \frac{L^3}{(2\pi)^3}\int d^3k, \tag{6.74}$$

leading to

$$\Im f(\mathbf{k},\mathbf{k}) = \frac{mL^3}{2\hbar^2}\left(\frac{2\pi\hbar^2}{mL^3}\right)^2\sum_{\mathbf{k}'}\left|f(\mathbf{k}',\mathbf{k})\right|^2\delta_{E,\frac{\hbar^2\mathbf{k}'^2}{2m}} \tag{6.75a}$$
$$\to \frac{2\pi^2\hbar^2}{mL^3}\frac{L^3}{(2\pi)^3}\int d^3k'\left|f(\mathbf{k}',\mathbf{k})\right|^2\delta\left(E-\frac{\hbar^2\mathbf{k}'^2}{2m}\right) \tag{6.75b}$$
$$= \frac{\hbar^2}{4\pi m}\int d^3k'\frac{d\sigma}{d\Omega_{k'}}\delta\left(E-\frac{\hbar^2\mathbf{k}'^2}{2m}\right). \tag{6.75c}$$

We have the volume element $d^3k' = k'^2dk'd\Omega_{k'}$, so we have

$$\Im f(\mathbf{k},\mathbf{k}) = \frac{\hbar^2}{4\pi m}\int d\Omega_{k'}\frac{d\sigma}{d\Omega_{k'}}\int dk'k'^2\delta\left(E-\frac{\hbar^2\mathbf{k}'^2}{2m}\right) \tag{6.76a}$$
$$= \frac{\hbar^2}{4\pi m}\frac{m}{k\hbar^2}\int d\sigma\int dk'k'^2\delta(k'-k) \tag{6.76b}$$
$$= \frac{k}{4\pi}\sigma_{\text{tot}}, \tag{6.76c}$$

where we have used the identity for the delta function

$$\delta(g(x)) = \frac{\delta(x-x_0)}{|\partial_x g(x_0)|}. \tag{6.77}$$

## 6.3 The Born Approximation

To calculate the scattering amplitude $f(\mathbf{k}',\mathbf{k})$ for some given potential $V(\mathbf{x})$ is done by calculating the matrix elements

$$\big\langle\mathbf{k}'\big|V\big|\psi^{(+)}\big\rangle = \langle\mathbf{k}'|T|\mathbf{k}\rangle \tag{6.78}$$

which is not easy as we lack the necessary closed analytical expressions.

We will consider the expansion of $T$ as powers of $V$, shown by the operator equation Eq. (6.32), which reads

$$T = V + V\frac{1}{E - H_0 - i\varepsilon}V + V\frac{1}{E - H_0 - i\varepsilon}V\frac{1}{E - H_0 - i\varepsilon}V + \cdots . \tag{6.79}$$

If we make the first order Born approximation, i.e. the first term, we get $T = V$, or equivalently $\left|\psi^{(+)}\right\rangle = |\mathbf{k}\rangle$. This gives

$$\begin{aligned} f^{(1)}(\mathbf{k}', \mathbf{k}) &= -\frac{mL^3}{2\pi\hbar^2}\langle\mathbf{k}'|V|\psi^{+}\rangle && (6.80\text{a}) \\ &= -\frac{mL^3}{2\pi\hbar^2}\int d^3x'\langle\mathbf{k}'|\mathbf{x}'\rangle\langle\mathbf{x}'|V|\mathbf{k}\rangle && (6.80\text{b}) \\ &= -\frac{mL^3}{2\pi\hbar^2}\int d^3x' V(\mathbf{x}')\langle\mathbf{k}'|\mathbf{x}'\rangle\langle\mathbf{x}'|\mathbf{k}\rangle && (6.80\text{c}) \\ &= -\frac{mL^3}{2\pi\hbar^2}\int d^3x' V(\mathbf{x}')\frac{e^{-i\mathbf{k}'\cdot\mathbf{x}'}}{L^{3/2}}\frac{e^{i\mathbf{k}\cdot\mathbf{x}'}}{L^{3/2}} && (6.80\text{d}) \\ &= -\frac{m}{2\pi\hbar^2}\int d^3x' V(\mathbf{x}')e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}'}. && (6.80\text{e}) \end{aligned}$$

Note that this is, disregarding the prefactor, the three-dimensional Fourier transform of the potential $V$ with respect to $\mathbf{q} = \mathbf{k} - \mathbf{k}'$.

Consider the important case where $V$ is a spherically symmetric potential, implying that $f^{(1)}(\mathbf{k}', \mathbf{k})$ is a function of $q = |\mathbf{q}|$. Since energy conservation implies $|\mathbf{k}'| = k$, we can write

$$q = 2k\sin\frac{\vartheta}{2}. \tag{6.81}$$

Using this in the integral we can evaluate by switching to spherical coordinates and letting the $z$-axis align with $\mathbf{q}$,

$$\begin{aligned} f^{(1)}(\vartheta) &= -\frac{m}{2\pi\hbar^2}\int d^3x' V(r)e^{iqr\cos\vartheta} && (6.82\text{a}) \\ &= -\frac{m}{2\pi\hbar^2}\int_0^\infty dr r^2 V(r)\underbrace{\int_0^\pi d\vartheta'\sin\vartheta' e^{iqr\cos\vartheta'}}_{u=\cos\vartheta',\ du=-\sin\vartheta d\vartheta'}\int_0^{2\pi} d\varphi && (6.82\text{b}) \\ &= -\frac{m}{\hbar^2}\int_0^\infty dr r^2 V(r)\int_{-1}^1 du e^{iqru} && (6.82\text{c}) \\ &= -\frac{m}{\hbar^2}\int_0^\infty dr r^2 V(r)\frac{e^{iqr} - e^{-iqr}}{iqr} && (6.82\text{d}) \\ &= -\frac{2m}{q\hbar^2}\int_0^\infty dr r V(r)\sin(qr) && (6.82\text{e}) \end{aligned}$$

We now consider scattering by a finite square well

$$V(r) = \begin{cases} V_0 & r \leq a \\ 0 & r > a. \end{cases} \tag{6.83}$$

The integral becomes

$$f^{(1)}(\vartheta) = -\frac{2m}{q\hbar^2}V_0\int_0^a dr r\sin(qr) \tag{6.84a}$$

$$= -\frac{2mV_0}{\hbar^2}\frac{1}{q}\left(\left[r\frac{-\cos(qr)}{q}\right]_0^a + \frac{1}{q}\int_0^a dr\cos(qr)\right) \tag{6.84b}$$

$$= -\frac{2mV_0}{\hbar^2}\frac{1}{q}\left(-a\frac{\cos(qr)}{q} + \frac{\sin(qr)}{q^2}\right) \tag{6.84c}$$

$$= -\frac{2mV_0}{\hbar^2}\frac{a^3}{(qa)^2}\left(\frac{\sin(qr)}{qa} - \cos(qa)\right). \tag{6.84d}$$

A nuclear potential is rather well approximated by a square well.

Considering another spherically symmetric potential, the Yukawa potential

$$V(r) = \frac{V_0 e^{-\mu r}}{\mu r} \tag{6.85}$$

where $V_0$ is independent of $r$ and $1/\mu$ is related to the range of the potential. The scattering amplitude is then

$$f^{(1)}(\vartheta) = -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\int_0^\infty dr r\frac{e^{-\mu r}}{r}\sin(qr) \tag{6.86a}$$

$$= -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\int_0^\infty dr e^{-\mu r}\sin(qr) \tag{6.86b}$$

$$= -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\frac{1}{2i}\int_0^\infty dr\left(e^{-(\mu-iq)r} - e^{-(\mu+iq)r}\right) \tag{6.86c}$$

$$= -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\frac{1}{2i}\left(\left[-\frac{e^{-(\mu-iq)r}}{\mu-iq}\right]_0^\infty - \left[-\frac{e^{-(\mu+iq)r}}{\mu+iq}\right]_0^\infty\right) \tag{6.86d}$$

$$= -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\frac{1}{2i}\left(\frac{1}{\mu-iq} - \frac{1}{\mu+iq}\right) \tag{6.86e}$$

$$= -\frac{2m}{\hbar^2}\frac{V_0}{\mu q}\frac{q}{\mu^2+q^2} \tag{6.86f}$$

$$= -\left(\frac{2mV_0}{\mu\hbar^2}\right)\frac{1}{q^2+\mu^2}. \tag{6.86g}$$

Note that

$$q^2 = 4k^2\sin\frac{\vartheta}{2} = 2k^2(1-\cos\vartheta). \tag{6.87}$$

The differential cross section for the Yukawa potential is

$$\frac{d\sigma}{d\Omega} = \left(\frac{2mV_0}{\mu\hbar^2}\right)^2\frac{1}{[2k^2(1-\cos\vartheta)+\mu^2]^2}. \tag{6.88}$$

If the ratio $V_0/\mu$ is fixed as we take the limit $\mu \to 0$, we find that we obtain the Coulomb potential. If we have $V_0/\mu = ZZ'e^2$ as $\mu \to 0$, we find

$$\frac{d\sigma}{d\Omega} = \frac{(2m)^2(ZZ'e^2)^2}{\hbar^4}\frac{1}{16k^4\sin^4(\vartheta/2)}, \tag{6.89}$$

and if we identify $|\mathbf{p}| = \hbar k$ and the kinetic energy $E_{\mathrm{KE}} = |\mathbf{p}|/2m$, we further find that

$$\frac{d\sigma}{d\Omega} = \frac{1}{16}\left(\frac{ZZ'e^2}{E_{\mathrm{KE}}}\right)^2 \frac{1}{\sin^4(\vartheta/2)}. \tag{6.90}$$

This equation is exactly the Rutherford scattering cross section which is obtained classically.

In general, if the scattering cross section for a spherically symmetric potential can be approximated with the first Born approximation, we have the following properties:

1. $d\sigma/d\Omega$ and $f(\vartheta)$, is a function of $q$ only, with $f(\vartheta)$ depending on the energy and scattering angle, only through $2k^2(1-\cos(\vartheta))$.

2. $f(\vartheta)$ is always real.

3. $d\sigma/d\Omega$ is independent of the sign of $V$.

4. For small $k$ ($q$ necessarily small)

$$f^{(1)}(\vartheta) = -\frac{1}{4\pi}\frac{2m}{\hbar^2}\int d^3x V(r) \tag{6.91}$$

   involving a volume integral independent of $\vartheta$. (Unsure about the factor $1/4\pi$)

5. $f(\vartheta)$ is small for large $q$ due to rapid oscillation of the integrand.

To investigate the validity of the Born approximation we return to the wave function

$$\langle \mathbf{x}|\psi^{(+)}\rangle = \langle \mathbf{x}|\mathbf{k}\rangle - \frac{2m}{\hbar^2}\int d^3x' \frac{e^{ik|\mathbf{x}-\mathbf{x}'|}}{4\pi|\mathbf{x}-\mathbf{x}'|} V(\mathbf{x}')\langle \mathbf{x}'|\psi^{(+)}\rangle. \tag{6.92}$$

The approximation is that $T \approx V$, or equivalently $\left|\psi^{(+)}\right\rangle = |\mathbf{k}\rangle$. The second term on the right-hand side must be much smaller than the first. Assuming a typical value for the potential to be $V_0$, and it is acting within some range $a$. Writing $r' = |\mathbf{x}-\mathbf{x}'|$ and evaluating the integral roughly, we obtain the validity condition

$$\left|\frac{2m}{\hbar}(\frac{4\pi}{3}a^3)\frac{e^{ikr'}}{4\pi a}V_0\frac{e^{i\mathbf{k}\cdot\mathbf{x}'}}{L^{3/2}}\right| \ll \left|\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{L^{3/2}}\right|. \tag{6.93}$$

In the low energy limit ($ka \ll 1$), the exponential factors may be replaced by unity, and ignoring factors of order unity, we get the simpler criterion

$$\frac{m|V_0|a^2}{\hbar^2} \ll 1. \tag{6.94}$$

Consider the Yukawa potential, with the range $a = 1/\mu$. The validity condition becomes

$$\frac{m|V_0|}{\hbar^2\mu^2} \ll 1. \tag{6.95}$$

Comparing this to the condition for the Yukawa potential to develop a bound state, which can be shown is

$$\frac{2m|V_0|}{\hbar^2\mu^2} \geq 2.7, \tag{6.96}$$

where $V_0$ is negative. It is clear that for a potential strong enough to have bound states, we cannot use the first order Born approximation. If we instead consider high energies ($ka \gg 1$), the exponential factors instead oscillate strongly over the integration, and cannot be set to unity. It is possible to show that, the validity condition is

$$\frac{2m}{\hbar^2}\frac{|V_0|a}{k}\ln(ka) \ll 1. \tag{6.97}$$

For larger $k$, the condition is more easily satisfied. Generally, the Born approximation tends to be better at higher energies.

### 6.3.1 The Higher-Order Born Approximation

Writing $T$ to the second order in $V$, we obtain

$$T = V + V\frac{1}{E - H_0 + i\varepsilon}V. \tag{6.98}$$

We continue the Born approximation with

$$f(\mathbf{k}', \mathbf{k}) \approx f^{(1)}(\mathbf{k}', \mathbf{k}) + f^{(2)}(\mathbf{k}', \mathbf{k}) \tag{6.99}$$

where we have

$$f^{(1)}(\mathbf{k}', \mathbf{k}) = -\frac{m}{2\pi\hbar^2}\int d^3x' V(\mathbf{x}')e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}'}, \tag{6.100}$$

and

$$f^{(2)}(\mathbf{k}', \mathbf{k}) = -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\int d^3x' \int d^3x'' \tag{6.101a}$$

$$\langle\mathbf{k}'|\mathbf{x}'\rangle V(\mathbf{x}')\left\langle\mathbf{x}'\middle|\frac{1}{E - H_0 + i\varepsilon}\middle|\mathbf{x}''\right\rangle V(\mathbf{x}'')\langle\mathbf{x}''|\mathbf{k}\rangle \tag{6.101b}$$

$$= -\frac{1}{4\pi}\frac{2m}{\hbar^2}\int d^3x' \int d^3x'' \tag{6.101c}$$

$$e^{-i\mathbf{k}'\cdot\mathbf{x}'}V(\mathbf{x}')\left[\frac{2m}{\hbar^2}G_+(\mathbf{x}', \mathbf{x}'')\right]V(\mathbf{x}'')e^{i\mathbf{k}\cdot\mathbf{x}''}. \tag{6.101d}$$

The physical interpretation of the second order is that the incident wave interacts with at $\mathbf{x}''$ via $V(\mathbf{x}'')$, then propagates from $\mathbf{x}''$ to $\mathbf{x}'$ via $G_+(\mathbf{x}', \mathbf{x}'')$, where another interaction occurs via $V(\mathbf{x}')$. Finally, the wave gets scattered in the direction $\mathbf{k}'$. We see that the second order is interpreted as a two-step scattering process. We can view the subsequent orders similarly.

## 6.4 Phase Shifts and Partial Waves

### 6.4.1 Free-Particle States

The Hamiltonian for a free particle is only the kinetic energy operator, which commutes with the momentum operator, and $\mathbf{L}^2$, and $L_z$. We can consider a simultaneous eigenket of $H_0, \mathbf{L}^2, L_z$, ignoring spin, called the spherical wave state, denoted $|E, l, m\rangle$. We can denote a general free-particle state as a superposition of $|E, l, m\rangle$ states, in the same way we might regard it as a superposition of $|\mathbf{k}\rangle$, with different $\mathbf{k}$. Thus, we can analyse a free-particle state, either using a plane wave basis $\{|\mathbf{k}\rangle\}$, or a spherical wave basis $\{|E, l, m\rangle\}$.

We want to find the transformation function $\langle \mathbf{k}|E,l,m\rangle$ between these two bases. First consider the normalization for spherical waves

$$\langle E',l',m'|E,l,m\rangle = \delta_{ll'}\delta_{mm'}\delta(E-E'). \tag{6.102}$$

We guess the angular dependence to be the same as for position-space wave functions, that is, spherical harmonics, and write

$$\langle \mathbf{k}|E,l,m\rangle = g_{lE}(k)Y_l^m(\hat{\mathbf{k}}), \tag{6.103}$$

where the function $g_{lE}(k)$, will be considered later. To prove this we start by considering a plane wave state propagating in the positive $z$-direction, $|k\hat{\mathbf{z}}\rangle$. This state has the important property

$$L_z|k\hat{\mathbf{z}}\rangle = (xp_y - yp_x)|k_x = 0, k_y = 0, k_z = k\rangle = 0. \tag{6.104}$$

Classically, this is reasonable since $\mathbf{L}\cdot\mathbf{p} = (\mathbf{x}\times\mathbf{p})\cdot\mathbf{p} = 0$. We can expand $|k\hat{\mathbf{z}}\rangle$ as

$$|k\hat{\mathbf{z}}\rangle = \sum_{l'}\int dE'|E',l',m'=0\rangle\langle E',l'm'=0|k\hat{\mathbf{z}}\rangle. \tag{6.105}$$

We do not sum over $m'$ since $\langle E',l',m'|k\hat{\mathbf{z}}\rangle = 0$ for $m' \neq 0$. To obtain a general ket $|\mathbf{k}\rangle$, we only have to apply the relevant rotation operator

$$|\mathbf{k}\rangle = \mathscr{D}(\alpha=\varphi,\beta=\vartheta,\gamma=0)|k\hat{\mathbf{z}}\rangle \tag{6.106}$$

Multiplying by $\langle E,l,m|$ from the left we obtain

$$\langle E,l,m|\mathbf{k}\rangle = \sum_{l'}\int dE'\langle E,l,m|\mathscr{D}(\alpha=\varphi,\beta=\vartheta,\gamma=0)|E',l'm'=0\rangle \tag{6.107a}$$

$$\times\langle E',l',m'=0|k\hat{\mathbf{z}}\rangle \tag{6.107b}$$

$$= \sum_{l'}\int dE'\mathscr{D}^{(l')}_{m0}(\alpha=\varphi,\beta=\vartheta,\gamma=0)\delta_{ll'}\delta(E-E') \tag{6.107c}$$

$$\times\langle E',l',m'=0|k\hat{\mathbf{z}}\rangle \tag{6.107d}$$

$$= \mathscr{D}^{(l)}_{m0}(\alpha=\varphi,\beta=\vartheta,\gamma=0)\langle E,l,m=0|k\hat{\mathbf{z}}\rangle. \tag{6.107e}$$

Since $\langle E,l,m=0|k\hat{\mathbf{z}}\rangle$ is independent of the orientation of $\mathbf{k}$, that it is independent of the angles $\vartheta,\varphi$, we can call it

$$\langle E,l,m=0|k\hat{\mathbf{z}}\rangle = \sqrt{\frac{2l+1}{4\pi}}g^*_{lE}(k). \tag{6.108}$$

We recall that we can write a spherical harmonic as

$$Y_l^{m'*}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\mathscr{D}^{(l)}_{m'0}(\alpha=\varphi,\beta=\vartheta,\gamma=0), \tag{6.109}$$

and we can then write

$$\langle \mathbf{k}|E,l,m\rangle = g_{lE}(k)Y_l^m(\hat{\mathbf{k}}). \tag{6.110}$$

We note

$$(H_0 - E)|E,l,m\rangle = 0, \tag{6.111}$$

and

$$\langle \mathbf{k}|(H_0 - E) = \left(\frac{\hbar^2 k^2}{2m} - E\right)\langle \mathbf{k}|. \tag{6.112}$$

Combining these we see that

$$\left(\frac{\hbar^2 k^2}{2m} - E\right)\langle \mathbf{k}|E, l, m\rangle = 0, \tag{6.113}$$

so $\langle \mathbf{k}|E, l, m\rangle$ can only be non-zero only if $E = \hbar^2 k^2/2m$, so we are able to write

$$g_{lE}(k) = N\delta\left(\frac{\hbar^2 k^2}{2m} - E\right). \tag{6.114}$$

Returning to the normalization condition we find

$$\langle E', l', m'|E, l, m\rangle = |N|^2 \frac{mk'}{\hbar^2}\delta(E - E')\delta ll'\delta(mm'), \tag{6.115}$$

where we see that $N = \hbar/\sqrt{mk}$ is sufficient. We therefore end up with

$$\langle \mathbf{k}|E, l, m\rangle = \frac{\hbar}{\sqrt{mk}}\delta\left(\frac{\hbar^2 k^2}{2m} - E\right)Y_l^m(\hat{\mathbf{k}}). \tag{6.116}$$

From this we can find the expansion of $|\mathbf{k}\rangle$ in the spherical wave states as

$$|\mathbf{k}\rangle = \sum_l \sum_m \int dE|E, l, m\rangle\langle E, l, m|\mathbf{k}\rangle \tag{6.117a}$$

$$= \sum_{l=0}^{\infty}\sum_{m=-l}^{l}\left|\frac{\hbar^2 k^2}{2m}, l, m\right\rangle\left(\frac{\hbar}{\sqrt{mk}}Y_l^{m*}(\hat{\mathbf{k}})\right) \tag{6.117b}$$

We now consider the wave function in position space, which from wave mechanics is known as

$$\langle \mathbf{x}|E, l, m\rangle = c_l j_l(kr)Y_l^m(\hat{\mathbf{r}}), \tag{6.118}$$

where $j_l(kr)$ is the spherical Bessel function of order $l$. To find $c_l$, we evaluate

$$\langle \mathbf{x}|\mathbf{k}\rangle = \sum_l \sum_m \int dE\langle \mathbf{x}|E, l, m\rangle\langle E, l, m|\mathbf{k}\rangle \tag{6.119a}$$

$$= \sum_l \sum_m \int dE c_l j_l(kr)Y_l^m(\hat{\mathbf{r}})\frac{\hbar}{\sqrt{mk}}\delta(E - \frac{\hbar^2 k^2}{2m})Y_l^{m*}(\hat{\mathbf{k}}) \tag{6.119b}$$

$$= \sum_l \frac{2l+1}{4\pi}P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}})\frac{\hbar}{\sqrt{mk}}c_l j_l(kr), \tag{6.119c}$$

where the addition theorem

$$\sum_m Y_l^m(\hat{\mathbf{r}})Y_l^{m*}(\hat{\mathbf{k}}) = \frac{2l+1}{4\pi}P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}) \tag{6.120}$$

have been used. We can also write

$$\langle \mathbf{x}|\mathbf{k}\rangle = \frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{(2\pi)^{3/2}} = \frac{1}{(2\pi)^{3/2}}\sum_l (2l+1)i^l j_l(kr)P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}), \tag{6.121}$$

which can be shown using the integral representation of the spherical Bessel function

$$j_l(kr) = \frac{1}{2i^l}\int_{-1}^{1} d(\cos\vartheta)e^{ikr\cos\vartheta}P_l(\cos\vartheta). \tag{6.122}$$

Comparing these two results gives

$$c_l = \frac{i^l}{\hbar}\sqrt{\frac{2mk}{\pi}}. \tag{6.123}$$

Thus, the wave functions are

$$\langle \mathbf{k}|E,l,m\rangle = \frac{\hbar}{\sqrt{mk}}\delta\left(\frac{\hbar^2k^2}{2m} - E\right)Y_l^m(\hat{\mathbf{k}}) \tag{6.124a}$$

$$\langle \mathbf{x}|E,l,m\rangle = \frac{i^l}{\hbar}\sqrt{\frac{2mk}{\pi}}j_l(kr)Y_l^m(\hat{\mathbf{r}}), \tag{6.124b}$$

which are very useful when developing partial-wave expansion.

### 6.4.2 Partial-Wave Expansion

We return to the case of $V \neq 0$, and assume a spherically symmetric potential. Then, the transition operator $T$ is a scalar operator that commutes with $\mathbf{L}^2$ and $\mathbf{L}$.

Using the spherical wave basis, the Wigner-Eckart theorem gives

$$\langle E',l',m'|T|E,l,m\rangle = T_l(E)\delta_{ll'}\delta_{mm'}, \tag{6.125}$$

where we see that $T$ is diagonal in both $l$ and $m$, and that the diagonal element depends on $E$ and $l$ but not $m$.

Considering the scattering amplitude

$$f(\mathbf{k}',\mathbf{k}) = -\frac{1}{4\pi}\frac{2m}{\hbar^2}L^3\langle\mathbf{k}'|T|\mathbf{k}\rangle \tag{6.126a}$$

$$\to -\frac{1}{4\pi}\frac{2m}{\hbar}(2\pi)^3\sum_l\sum_m\sum_{l'}\sum_{m'}\int dE\int dE' \tag{6.126b}$$

$$\times\langle\mathbf{k}'|E',l',m'\rangle\langle E',l',m'|T|E,l,m\rangle\langle E,l,m|\mathbf{k}\rangle \tag{6.126c}$$

$$= -\frac{1}{4\pi}\frac{2m}{\hbar^2}(2\pi)^3\frac{\hbar^2}{mk}\sum_l\sum_m T_l\left(\frac{\hbar^2k^2}{2m}\right)Y_L^m(\hat{\mathbf{k}}')Y_l^{m*}(\hat{\mathbf{k}}) \tag{6.126d}$$

$$= -\frac{4\pi^2}{k}\sum_l\sum_m T_l\left(\frac{\hbar^2k^2}{2m}\right)Y_l^m(\hat{\mathbf{k}}')Y_l^{m*}(\hat{\mathbf{k}}). \tag{6.126e}$$

Choosing a coordinate system such that $\mathbf{k}$ is in the positive $z$-direction, we have

$$Y_l^m(\hat{\mathbf{k}}) = \sqrt{\frac{2l+1}{4\pi}}\delta_{m0}. \tag{6.127}$$

Taking $\vartheta$ to be the angle between $\mathbf{k}$ and $\mathbf{k}'$, we can write

$$Y_l^0(\mathbf{k}') = \sqrt{\frac{2l+1}{4\pi}}P_l(\cos\vartheta). \tag{6.128}$$

We define a partial-wave amplitude $f_l(k)$ as

$$f_l(k) = -\frac{\pi T_l(E)}{k}. \tag{6.129}$$

We can now rewrite the scattering amplitude to

$$f(\mathbf{k}', \mathbf{k}) = f(\vartheta) = \sum_{l=0}^{\infty}(2l+1) f_l(k) P_l(\cos\vartheta). \tag{6.130}$$

Studying the large-distance behaviour of the wave function $\langle \mathbf{x} | \psi^{(+)} \rangle$, using what is found above, gives

$$\langle \mathbf{x} | \psi^{(+)} \rangle = \frac{1}{(2\pi)^{3/2}} \sum_l (2l+1) \frac{P_l}{2ik} \left[ (1 + 2ikf_l(k)) \frac{e^{ikr}}{r} - \frac{e^{-i(kr - l\pi)}}{r} \right]. \tag{6.131}$$

We can now interpret the physics of scattering. When a scatterer is absent, we can model the plane wave as a sum of spherically incoming and outgoing waves. The presence of a scatter changes the coefficient of the outgoing waves by

$$1 \to 1 + 2ikf_l(k), \tag{6.132}$$

leaving the incoming wave unaffected.

### 6.4.3 Unitarity and Phase Shifts

We will now investigate the consequences of unitarity, or equivalently probability conservation. For a time independent formulation the flux current density $\mathbf{j}$ must satisfy

$$\nabla \cdot \mathbf{j} = -\partial_t |\psi|^2 = 0. \tag{6.133}$$

Consider a spherical surface of large radius, then by Gauss' theorem we have

$$\int_S \mathbf{j} \cdot d\mathbf{S} = 0. \tag{6.134}$$

Physically, these two equations tells us that there are no sources or sinks of particles, and the outgoing and incoming flux must be the same. Due to angular-momentum conservation this must hold for each partial wave separately, or in other words, we must have the coefficient of $e^{ikr}/r$ have the same magnitude as the coefficient of $e^{-ikr}/r$. We define $S_l(k)$ as

$$S_l(k) = 1 + 2ikf_l(k) \tag{6.135}$$

and this is then

$$|S_l(k)| = 1. \tag{6.136}$$

Thus, the only thing that can happen is a change of phase, not magnitude, of the outgoing wave. This equation is known as the unitarity relation for the $l$th partial wave. We can also regard $S_l(k)$ as the $l$th diagonal element of the $S$ operator, which has unitarity as a requirement due to probability conservation.

We can therefore at large distances, at most see a phase change of the outgoing wave in the wave function as a result of scattering. By convention, we call this phase $2\delta_l$, and we write

$$S_l = e^{2i\delta_l}, \tag{6.137}$$

where $\delta$ is real. We note that $\delta_l$ is an implicit function of $k$. We can now rewrite the partial-wave amplitude as

$$f_l = \frac{S_l - 1}{2ik}, \tag{6.138}$$

or in terms of $\delta_l$

$$f_l = \frac{e^{2i\delta_l} - 1}{2ik} = \frac{e^{i\delta_l}\sin\delta_l}{k} = \frac{1}{k\cot\delta_l - ik}, \tag{6.139}$$

depending on what is convenient. The full scattering amplitude is

$$f(\vartheta) = \sum_{l=0}(2l+1)\left(\frac{e^{2i\delta_l} - 1}{2ik}\right)P_l(\cos\vartheta) \tag{6.140a}$$

$$= \frac{1}{k}\sum_{l=0}(2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos\vartheta). \tag{6.140b}$$

This expression rests on the principles of rotational invariance and probability conservation. It is possible to obtain the same expression by solving the Schrödinger equation with a real, spherically symmetric potential. However, our treatment is of interest as it can more easily be generalized to situations where nonrelativistic potentials fails.

We find the total cross section by

$$\sigma_{\text{tot}} = \int d\Omega \frac{d\sigma}{d\Omega} = \int d\Omega |f(\vartheta)|^2 \tag{6.141a}$$

$$= \frac{1}{k^2}\int_0^{2\pi} d\varphi \int_{-1}^{1} d(\cos\vartheta)\sum_l\sum_{l'}(2l+1)(2l'+1)e^{i\delta_l}\sin\delta_l e^{-i\delta_{l'}}\sin\delta_{l'}P_lP_{l'} \tag{6.141b}$$

$$= \frac{4\pi}{k^2}\sum_l(2l+1)\sin^2\delta_l. \tag{6.141c}$$

As $\delta_l$ is a function of energy, so is $f_l(k)$. The partial wave amplitude, has the restriction of the unitarity relation, which we can explore by drawing an Argand diagram for $kf_l$, noting that

$$kf_l = \frac{i}{2} + \frac{1}{2}e^{-(i\pi/2)+2i\delta_l}. \tag{6.142}$$

We find a circle with radius 1/2 centred at $z = i/2$, called the unitary circle, on which $kf_l$ must lie. If $\delta_l$ is small we find

$$f_l = \frac{e^{i\delta_l}\sin\delta_l}{k} \approx \frac{(1+i\delta_l)\delta_l}{k} \approx \frac{\delta_l}{k}, \tag{6.143}$$

and $f_l$ is thus almost purely real. If we instead consider $\delta_l$ close to $\pi/2$ we have that $f_l$ is almost purely imaginary with maximized magnitude, and the $l$th partial wave can be considered to be in resonance. The maximum partial cross section is

$$\sigma_{\max}^{(l)} = \frac{4\pi}{k^2}(2l+1), \tag{6.144}$$

and is achieved when $\sin^2\delta_l = 1$.

### 6.4.5 Hard-Sphere Scattering

Consider scattering by a hard rigid sphere, that is, we have the potential

$$V = \begin{cases} \infty & r < R \\ 0 & r > R. \end{cases} \tag{6.145}$$

We know that the wave function must vanish at $r = R$ since the potential is impenetrable. We have

$$A_l(r)|_{r=R} = 0, \tag{6.146}$$

or

$$j_l(kR)\cos\delta_l - n_l(kR)\sin\delta_l = 0, \tag{6.147}$$

or

$$\tan\delta_l = \frac{j_l(kR)}{n_l(kR)}. \tag{6.148}$$

Thus, we know the phase shifts in terms of the spherical Bessel functions.

Considering the $l = 0$ case, called $S$-wave scattering, we find

$$\tan\delta_0 = \frac{\sin(kR)/kR}{\cos(kR)/kR} = -\tan kR, \tag{6.149}$$

or $\delta_0 = -kR$. We find the radial wave function as

$$A_0(r) \propto \frac{\sin kr}{kr}\cos\delta_0 + \frac{\cos kr}{kr}\sin\delta_0 = \frac{1}{kr}\sin(kr + \delta_0). \tag{6.150}$$

Plotting this gives a sinusoidal wave shifted compared to a free wave by an amount $R$. We want to study the low-energy ($kR$ small and $kR \ll 1$) and high-energy ($kR$ large) limits. For the low-energy limit, we use

$$j_l(kr) \approx \frac{(kr)^l}{(2l+1)!!} \tag{6.151a}$$

$$n_l(kr) \approx -\frac{(2l-1)!!}{(kr)^{l+1}} \tag{6.151b}$$

to obtain

$$\tan\delta_l = \frac{-(kR)^{2l+1}}{\{(2l+1)[(2l-1)!!]^2\}}. \tag{6.152}$$

We see that for $l > 0$ that the contribution goes to 0 quickly since $kR \ll 1$, and we have pure $S$-wave scattering. Because $\delta_0 = -kR$ is no matter the size of $k$ we find the differential cross section

$$\frac{d\sigma}{d\Omega} = \frac{\sin^2\delta_0}{k^2} =\approx R^2 \tag{6.153}$$

and the total cross section is

$$\sigma_{\text{tot}} = \int d\Omega \frac{d\sigma}{d\Omega} = 4\pi R^2. \tag{6.154}$$

Interestingly, this is exactly 4 times larger than the geometric cross section, the disc of radius $R$ obtained by the shadows blocking the propagating plane wave.

## 6.6 Low-Energy Scattering and Bound States

When $\lambda = 1/k$ is comparable to or larger than the potential range $R$, (low energy regime), partial waves for $l > 0$ are generally unimportant. The phase-shift goes to zero as

$$\delta l \sim k^{2l+1}. \tag{6.155}$$

### 6.6.1 Rectangular Well or Barrier

We will consider $S$-wave scattering

$$V = \begin{cases} V_0 = \text{constant} & \text{for } r < R \\ 0 & \text{otherwise} \end{cases} \begin{cases} V_0 > 0 & \text{repulsive} \\ V_0 < 0 & \text{attractive.} \end{cases} \tag{6.156}$$

Many of the features obtained by considering this potential are applicable to other finite range potentials. We have already discussed the outside solution. The inside solution may be obtained for a constant $V_0$ as

$$u = rA_{l=0}(r) \propto \sin(k'r), \tag{6.157}$$

where $k'$ is determined by

$$E - V_0 = \frac{\hbar^2 k'^2}{2m}, \tag{6.158}$$

using the boundary condition $u(r = 0) = 0$. That is, we get a sinusoidal function inside the potential as well, as long as $E > V_0$. The curvature is however different as it depends on if the potential is attractive or repulsive. If we instead have $V_0 > E$ we get

$$u(r) \propto \sinh(\kappa r), \tag{6.159}$$

with $\kappa$ defined by

$$\frac{\hbar\kappa^2}{2m} = V_0 - E. \tag{6.160}$$

We consider the attractive case ($V_0 < 0$), and imagine the magnitude to increase, leading to increased attraction. This results in a wave function with even larger curvature. If we suppose that the depth of the well is such that the interval $[0, R]$ fits one quarter of a cycle of the sinusoidal wave, and using the low-energy $kR \ll 1$ limit, we find that the phase shift is $\delta_0 = \pi/2$. This results in maximum $S$-wave cross section for a given $k$. Increasing the magnitude of $V_0$ until only a half-cycle fits, we find the phase shift to be $\delta_0 = \pi$. Remarkably, the partial-wave cross section

$$\sigma_{l=0} = 0 \tag{6.161}$$

vanishes, despite the strong attraction. If the energy is low enough such that $l \neq 0$ does not contribute we get an almost perfect transmission of the wave incident on the potential. This effect is called Ramsauer-Townsend effect, and is observed experimentally for electron scattering by noble gases and was discovered in 1923 before the birth of wave mechanics.

# 7 Identical Particles

# 8 Relativistic Quantum Mechanics